Software Requirements Specification for FitFolio

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1 Introduction

This document is the Software Requirements Specification (SRS) for FitFolio, a mobile application in which young adults earn virtual currency from the physical activity their phone records and invest it in a simulated market that follows real-world prices. It states what FitFolio must do and the qualities it must have, without prescribing how it will be built. The SRS is the reference against which the development team and the client agree on scope, the basis for the design, and the source from which the Verification and Validation (V&V) Plan derives its tests. It also records the safety and security requirements identified in the Hazard Analysis (HA), so that design and testing treat them like every other requirement.

The structure of this document follows the course SRS template, which is based on ISO/IEC/IEEE 29148, the international standard for requirements engineering (ISO, 2018), and is consistent with the system life-cycle processes of ISO/IEC/IEEE 15288 (ISO, 2023b). This section states the purpose and scope of FitFolio, identifies its stakeholders, defines the terms used throughout the document, and describes the knowledge expected of its readers.

1.1 Purpose

FitFolio is intended to help young adults build fitness and financial habits together, at the point in life when starting either one pays off most. It rewards the physical activity users already do with virtual currency, lets them invest that currency in a simulated market at real-world prices, explains financial and fitness concepts where they appear, and lets them share the progress they choose with friends. Because the currency has no real-world value, users can learn by making investing decisions and seeing their results without risking money. The problem that motivates FitFolio is described in detail in the Problem Statement in Section 2.

The same habits also matter beyond the individual user. Canadian guidelines recommend that adults accumulate at least 150 minutes of moderate to vigorous physical activity each week (Canadian Society for Exercise Physiology, 2020), and the national financial literacy strategy aims for a Canada in which everyone can build financial resilience (Financial Consumer Agency of Canada, 2021). By making both habits approachable and low-risk, FitFolio is intended to contribute in a small way to both goals.

1.2 Scope

FitFolio is a single product: a smartphone application for Android and iOS, together with the server that supports it.

Problem being solved. Young adults put off building fitness and investing habits because investing advice is confusing and learning by doing risks real money, while beginners at exercise lose motivation before they see results.

Intended users. Adults aged 18 or older, most of them students or people in their first jobs, with little or no investing experience and any level of fitness (Section 2.4).

Major functionality. Within scope, FitFolio:

- reads activity records from the phone's health platform and converts them into virtual currency;
- lets users buy and sell shares of a curated list of real-world stocks and exchange-traded funds (ETFs) at prices that follow the real market;
- shows portfolio value, activity trends, and milestones over time;
- explains financial and fitness concepts where they appear, gives feedback on the user's own decisions, and tracks learning progress through knowledge checks;
- replays past market periods, such as a downturn, as practice scenarios that do not affect the user's live portfolio;
- shares the statistics and milestones that the user selects with accepted friends.

Expected benefits. Users practice investing with no financial risk, stay motivated to exercise because activity has a visible use, and learn core concepts by applying them to their own decisions. The targets that measure these benefits are stated as non-functional requirements.

Out of scope. FitFolio does not:

- handle real money: the currency cannot be bought, withdrawn, or exchanged, and FitFolio never connects to a bank or brokerage account;
- give financial advice: it never recommends buying or selling a real-world security;
- give medical or training advice: it does not diagnose conditions, prescribe exercise programs, or set diet or weight goals;
- integrate directly with wearable devices (a stretch goal), since all activity is read through the phone's health platform;
- offer every listed security, or options, futures, short selling, or borrowing to invest;
- update prices tick by tick: simulated prices follow real prices with a bounded delay, because market data providers limit how often prices can be requested;
- predict future prices: scenarios replay only historical data;
- support public posts, direct messages, or rankings between users;

- run on tablets, the web, or desktop computers.

These limits also bound the inputs FitFolio must handle. For example, a day's step count ranges from zero on a rest day to tens of thousands for a very active user, and every asset price is a positive amount reported by the market data provider.

Standards and regulations. The requirements are written to comply with the following external standards and regulations, which also define the formats of the data FitFolio exchanges:

- **Privacy:** the Personal Information Protection and Electronic Documents Act (PIPEDA) (Government of Canada, 2000), which governs the collection, use, and deletion of users' personal and health data.
- **Accessibility:** the Web Content Accessibility Guidelines (WCAG) 2.2 (World Wide Web Consortium, 2023), applied to the app's screens.
- **Security:** the OWASP Mobile Application Security Verification Standard (MASVS) (OWASP Foundation, 2024).
- **Quality characteristics:** the product quality model of ISO/IEC 25010 (ISO, 2023a), which names the categories used for the non-functional requirements.
- **Data formats:** JavaScript Object Notation (JSON) (Bray, 2017) for data exchanged between the app, the server, and the market data provider; ISO 8601 (ISO, 2019) for dates, times, and time zones; and ISO 4217 (ISO, 2015) for the currency in which each asset price is quoted.

1.3 Stakeholders

Table 1 lists the stakeholders of FitFolio. Primary stakeholders use the product or define its direction, secondary stakeholders are affected by it indirectly, and external stakeholders are organizations whose services and rules constrain it. Each stakeholder has an identifier, so that every stakeholder requirement in Section 3 can name the stakeholders it comes from.

Table 1: Stakeholders of FitFolio

ID	Stakeholder	Role and interest
Primary		
SH1	End users (beginner investors and fitness-motivated users)	Use FitFolio to earn currency from activity, practice investing, and learn the concepts involved. Want an engaging, low-risk experience that is easy to understand.
SH2	Customer and project sponsor (Chris Schankula)	Proposed the project and defines its vision of gamifying exercise and teaching investing in a low-stakes setting. Approves the scope and validates the product against that vision.
SH3	Development team (Team #1, Burn n' Earn)	Builds and maintains FitFolio. Needs requirements that are complete, unambiguous, and verifiable.
SH4	Administrators	Operate the server after release: manage the list of tradable assets, review educational content, and respond to outages. The development team fills this role; FitFolio has no separate administrator account.
Secondary		
SH5	Users as friends (a role of SH1)	View the updates a user chooses to share. Expect to see only what the user selected.
SH6	Society	Benefits if more young adults build lasting exercise and financial habits (Canadian Society for Exercise Physiology, 2020; Financial Consumer Agency of Canada, 2021). Is harmed if the app spreads financial misinformation, encourages unhealthy exercise, or exposes personal data.
External		
SH7	Health platform providers (Apple HealthKit, Google Health Connect)	Supply activity data. Their application programming interfaces (APIs), permission models, and policies limit what FitFolio can read.
SH8	Market data providers	Supply real-world prices. Their accuracy, request limits, and cost limit how closely the simulated market follows the real market.
SH9	App stores and privacy regulators	Set the rules for distributing an app that handles health data and financial content, including PIPEDA (Government of Canada, 2000).

1.4 Definitions and Acronyms

Table 2 defines the terms used in this document, in alphabetical order, followed by the acronyms. Each term is used only with the meaning given here. Where an acronym stands for a defined term, its entry links to that definition.

Table 2: Definitions and Acronyms

Term	Definition
Terms	
Activity goal	A daily amount of activity that the user chooses to aim for, such as a number of steps.
Activity record	A measured amount of one activity type over a stated time interval, as reported by the health platform; for example, 4,200 steps between 9:00 and 10:00 on a given day.
Activity type	A category of physical activity recorded by the health platform, such as steps, cycling or wheelchair pushes. The supported types are listed in `SUPPORTED_ACTIVITY_TYPES` (Appendix C).
Asset	A real-world security that can be traded in FitFolio: either a stock or an exchange-traded fund.
Balance	The amount of virtual currency a user holds and has not invested.
Concept explanation	A short description of a financial or fitness concept, shown where the concept appears in the app.
Diversification	The extent to which a portfolio's value is spread across different assets, so that a fall in one asset has a limited effect on the whole.
Exchange-traded fund	A fund that holds a collection of assets, such as the stocks of many companies, and is itself traded like a stock.
Feedback	An explanation, specific to the user's own portfolio or activity, of what happened and why; for example, why their portfolio's value fell on a given day.
Fitness milestone	A stated level of activity reached by the user, such as their first week with activity recorded every day.
Friend	Another FitFolio user who has accepted a user's friend request.
Health platform	The service built into a smartphone's operating system that stores health and activity data, such as Apple HealthKit on iOS and Google Health Connect on Android.
Holding	The number of shares of one asset that a user owns.
Knowledge check	A short set of questions that tests whether a user understands a concept.
Learning progress	The concepts a user has viewed and the knowledge checks they have passed.
Manually entered record	An activity record that the user typed into the health platform, rather than one measured by a sensor.

Table 2 continued

Term	Definition
Market data provider	An external service that reports current and historical market prices.
Market price	The most recent price of one share of an asset on the real-world exchange where it is listed.
Market scenario	A replay of the price history of a past period, such as a market downturn, in which the user trades with a portfolio separate from their live portfolio.
Portfolio	All of a user's holdings, together with the history of their trades.
Portfolio value	The total value of a user's holdings at their current simulated prices.
Price age	The time elapsed since a simulated price was last updated from the market data provider.
Raw health data	Individual activity records and body measurements (age, height, and weight), as opposed to summaries the user has chosen to share.
Reminder	A local notification from FitFolio of one of the types in Table 6.
Rest day	A day that the user marks as a planned break from activity; it does not end their streak.
Reward	An amount of virtual currency credited to a user for an activity record.
Share	One unit of an asset.
Shared update	A statistic or milestone that a user has chosen to make visible to their friends.
Simulated market	The set of assets that users can trade in FitFolio, together with their simulated prices.
Simulated price	The price at which an asset is traded in FitFolio. In the live market it follows the market price; in a market scenario it follows a historical price.
Social feed	The list of shared updates from a user's friends.
Stock	A share of ownership in a single public company.
Streak	The number of consecutive days that count toward the user's streak: days on which the user met every activity goal in effect, recorded any activity when no goal was set, or took a rest day (Section 6.7).
Symbol	A code identifying an asset on an exchange.
Trade	The purchase of shares of an asset for virtual currency, or the sale of shares for virtual currency.

Table 2 continued

Term	Definition
Virtual currency	The unit of value earned in FitFolio. It has no real-world monetary value and cannot be exchanged for money.
Workout	An activity record marked as a deliberate exercise session, with a start time, an end time, and an activity type.
Acronyms	
AI	Artificial Intelligence
API	Application Programming Interface
CA	Critical Assumption (Hazard Analysis identifier prefix)
ETF	Exchange-Traded Fund
FR	Functional Requirement
HA	Hazard Analysis
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LC	Likely Change
MASVS	Mobile Application Security Verification Standard
NFR	Non-Functional Requirement
OS	Operating System
OWASP	Open Worldwide Application Security Project
PIPEDA	Personal Information Protection and Electronic Documents Act
RRSP	Registered Retirement Savings Plan
SAF	Safety requirement (Hazard Analysis identifier prefix)
SC	System Constraint
SEC	Security requirement (Hazard Analysis identifier prefix)
SH	Stakeholder (identifier prefix in Table 1)
SR	Stakeholder Requirement
SRS	Software Requirements Specification
TFSA	Tax-Free Savings Account
UC	Use Case
ULC	Unlikely Change
UUID	Universally Unique Identifier
V&V	Verification and Validation
WCAG	Web Content Accessibility Guidelines

1.5 Characteristics of Intended Reader

The intended readers of this SRS are the FitFolio development team, the client, course reviewers, and any future maintainer: the people who will review the requirements and design software to meet them. The end users of FitFolio are a different group and are described in Section 2.4. A reader is expected to have:

- **Software engineering:** third-year undergraduate knowledge of software requirements, including functional and non-functional requirements, use cases, traceability, and verification, at the level of a course such as SFWRENG 3RA3 (Software Requirements and Security Considerations).
- **Mathematics:** high-school algebra and percentages, enough to follow how rewards and portfolio values are calculated.
- **Mobile applications:** everyday familiarity with smartphone applications, including installing an app and responding to permission prompts. No mobile development experience is needed.
- **Finance and fitness:** no prior knowledge. Every financial and fitness term the requirements rely on is defined in Table 2.

Because an SRS is a reference document rather than one read from start to finish, readers new to FitFolio are encouraged to begin with the Problem Statement and Product Vision in Section 2, then the Functional and Non-Functional Requirements, and to consult other sections as needed.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

Current situation. Building good fitness and financial habits early pays off for decades, because both compound over time. Regular exercise builds on itself, and so does money invested early. Poor habits compound in the same way. Yet many young adults, particularly students and those starting their first jobs, struggle to begin either one.

Problem. Advice on managing money is scattered across many sources, which often contradict each other, and learning to invest by doing is risky because beginners' mistakes cost real money. Starting to exercise is similarly daunting, and many beginners lose motivation before they see results.

Impact. Many people put off both habits until later, when poor habits are already established and harder to change.

Desired improvement. An approachable, low-risk way for beginners to build fitness and financial habits together while staying motivated.

2.2 Product Vision

FitFolio is a mobile app in which users earn virtual currency from their everyday physical activity and invest it in a simulated market that follows real-world prices. It lets beginners build fitness habits and learn to invest without risking real money. Its goals, which the requirements in this document discuss, are:

G1 (Learning by doing). Users learn core financial and fitness concepts through hands-on, low-risk practice, with each decision paired with feedback explaining what happened and why. *Measured by:* a before-and-after knowledge check on a defined set of core concepts, where at least 70% of usability test participants improve their score.

G2 (A human-centered, all-in-one experience). Users see their fitness progress and financial learning together in one place, so they rarely need to leave the app to understand what they are looking at. *Measured by:* usability testing, where at least 80% of first-time participants complete the core tasks without outside help.

G3 (Motivation through friends, with privacy). Users stay motivated by sharing their progress with friends, while keeping control of their data. *Measured by:* users control who sees each shared statistic, and no raw health data is shared unless the user selects it.

G4 (Risk-free investing practice). Users practice investing in a simulated market based on real-world market data, without risking real money. *Measured by:* users can trade at least 50 real-world stocks at prices within 15 minutes of the real market, and can practice in market scenarios such as a replay of a past market downturn.

G5 (Rewards for everyday activity). Users are rewarded for the activities they already do, as recorded by their phone's health platform. *Measured by:* at least five activity types are supported.

2.3 System Context

Figure 1 shows FitFolio as a single system, with its mobile app, server and database inside the boundary, and the external entities it exchanges information with. Activity sources, such as phone sensors, smartwatches and other tracker apps, record activity in the phone's health platform. They never communicate with FitFolio directly, so they are shown outside the boundary with a dashed outline. FitFolio does not connect to banks, brokerages, personal portfolio trackers, or the phone's location services or contacts. Reminders are generated on the phone as local notifications, so FitFolio uses no push notification service.

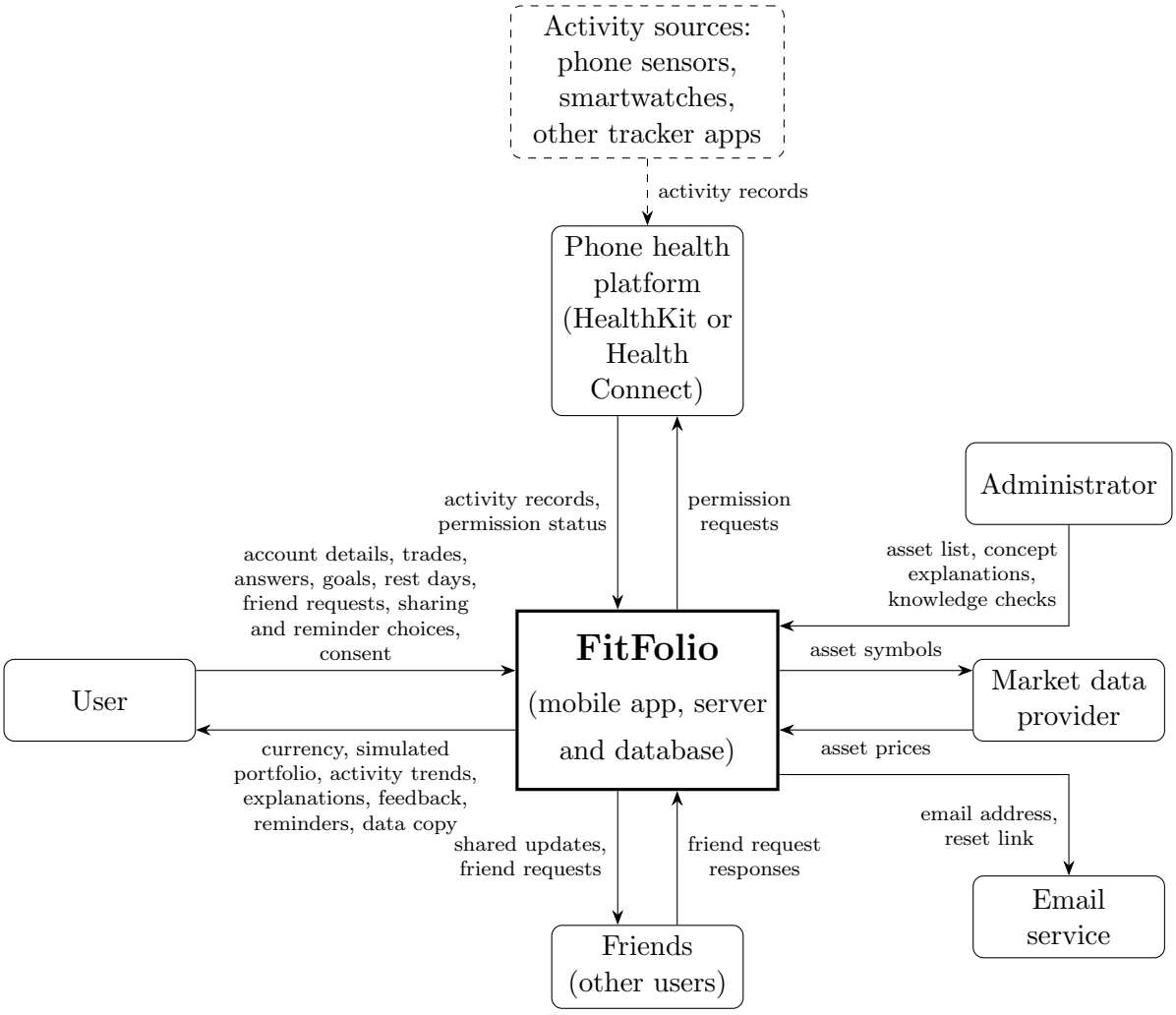


Figure 1: System context of FitFolio

The responsibilities of each entity in Figure 1 are:

- User Responsibilities:
  - Confirm truthfully that they are at least `MIN_USER_AGE` years old.
  - Keep any profile measurements they choose to enter, such as height and weight, accurate.
  - Grant FitFolio the permissions its features use, and keep an activity source recording to the phone's health platform.
  - Make their own trade decisions. FitFolio gives no financial advice.
  - Choose what to share, with which friends, and when to receive reminders.
  - Keep an Internet connection for trading and the social feed.
  - Keep their password private and keep access to their account's email address.
- FitFolio Responsibilities:
  - Award currency only for activity within plausibility limits and up to a daily cap (Hazard Analysis SAF-7, SEC-1).
  - Treat a day with no activity data as a day with zero activity.
  - Reject purchases that cost more than the user's balance and sales of more shares than the user holds (Hazard Analysis SEC-5).
  - Record each trade once and completely (Hazard Analysis SEC-5, SEC-6).
  - Show each asset price with the time it was obtained, and mark delayed prices (Hazard Analysis SEC-2).
  - Refuse trades on out-of-date prices while the market is open, and queue trades requested while it is closed (Hazard Analysis SEC-3, SEC-4).
  - Show each shared update only to the friends the user chose, without location detail (Hazard Analysis SEC-13, SEC-14).
  - Request only the permissions its features use, explaining each one (Hazard Analysis SEC-7, SEC-8).
  - Send the market data provider only the symbols of the assets being priced, and no personal or health data (Hazard Analysis SEC-11).
  - Send the email service no personal or health data other than the user's email address and their password reset link (Hazard Analysis SEC-11).
- Health Platform Responsibilities:
  - Store activity records from the user's activity sources, and provide the records for the activity types the user has permitted.

  - Grant or withdraw FitFolio's read access as the user chooses, and report the current permission status.
  - Keep the activity data it records as accurate as its activity sources allow.

- Market Data Provider Responsibilities:
  - Supply current and historical prices for the assets FitFolio requests, within its request limits.

- Email Service Responsibilities:
  - Deliver each password reset link that FitFolio sends to the user's email address.

- Administrator Responsibilities:
  - Keep the list of tradable assets current.
  - Keep concept explanations and knowledge checks accurate, and name no asset for the user to buy or sell in them (Hazard Analysis SAF-9).

- Friend Responsibilities:
  - Accept or decline a user's friend request.
  - View the updates a user has shared with them in their social feed.

FitFolio is intended for personal education and habit building. Members of the general public use it on their own smartphones to track their activity, earn currency and practice investing in a simulated market. It is not intended for professional, engineering or scientific use.

FitFolio is not mission-critical: it holds and moves no real money, so an outage only delays rewards and learning. It is not safety-critical either since it controls no physical equipment and provides no medical function. However, it handles personal health data and is designed to influence how users exercise and make financial decisions. These hazards, and the requirements that control them, are covered in the Hazard Analysis. The rationale in Section 11 therefore focuses on those requirements.

2.4 User Characteristics

The typical user of FitFolio is a young adult, such as a student or someone starting their first job (Problem Statement and Goals, Section 1). The user of FitFolio:

- Is at least `MIN_USER_AGE` years old (Hazard Analysis CA2).

- Uses a smartphone daily, and can install an app, grant or deny app permissions, and use common mobile app controls.

- Carries an Android or iOS phone during everyday activity. No wearable device or fitness expertise is required.
- Reads English, and has the numeracy of Grade 9 mathematics: working with percentages, comparing values, and reading a simple line chart.
- Needs no prior knowledge of investing terms such as stock, exchange-traded fund (ETF) or portfolio.

As per interviews conducted, participants mainly track their activity with a phone. Both already knew some investing concepts, through self-study or formal study, and both learned most by doing. No one interviewed was a self-reported complete beginner, so the beginner characteristics above come from the Problem Statement, and usability testing should include beginners.

2.5 System Constraints

The following constraints limit the design of FitFolio. Each one is imposed by the project brief, an external service, the law or the app stores.

SC1: FitFolio shall run as a mobile app on Android phones running Android `MIN_ANDROID_VERSION` or later and on iPhones running iOS `MIN_IOS_VERSION` or later. *Rationale:* This project calls for a mobile app, and these are the versions on the devices available to the team and the supervisor for testing (Development Plan, Section 10).

SC2: FitFolio shall obtain recorded activity only through the phone's health platform: Apple HealthKit on iOS and Health Connect on Android. This rule also covers any library FitFolio uses to read health data. No core feature shall require a wearable device. *Rationale:* Users carry their phone during everyday activity (Section 2.4), and the health platforms already collect data from the phone and from any wearable the user owns (note, wearable integration is a stretch goal).

SC3: FitFolio shall access health data only under the access rules Apple (iOS provider) and Google (Android OS provider) set for their health platforms, such as declaring each data type it reads and explaining each permission to the user (Apple Inc., 2026; Google LLC, 2026). *Rationale:* Apps that break these rules are not accepted in the app stores. The rules also limit what FitFolio can read, such as how far back past activity data can be read.

SC4: FitFolio shall obtain real-world asset prices from a third-party market data provider (`MARKET_DATA_PROVIDER`), within the provider's request limits and licence terms. *Rationale:* The team cannot produce market prices itself, so request limits, cost and permitted use are set by the provider.

SC5: Virtual currency and simulated holdings shall have no monetary value. They cannot be bought with, redeemed for or exchanged for money, and FitFolio shall not connect to any bank or brokerage account or give personalized investment advice. *Rationale:* Handling real money or giving personalized investment advice would bring FitFolio under financial regulation that the team is not licensed for. Keeping real money out also keeps the practice risk-free (G4).

SC6: FitFolio shall collect, use, store and delete personal and health data in line with Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) and the health-data policies of the Apple App Store and Google Play. *Rationale:* These laws and policies apply to any app that handles this data in Canada and in these stores. The specific controls are in Section 7.3 and the Hazard Analysis (SEC-7 to SEC-12 and SEC-18).

SC7: Trading, market prices and the social feed shall require an Internet connection to FitFolio's server. *Rationale:* Prices are fetched from the provider once for all users to stay within its request limits (SC4), friends' shared updates must reach other users' phones, and a user's portfolio must survive the loss of their phone.

3 Stakeholder Requirements

This section lists what the stakeholders in Table 1 need from FitFolio, stated without reference to any technology. Each requirement names its *source* stakeholders, the *evidence* for it, and the product *goal* from Section 2 that it supports. Use cases in Section 5 and functional requirements in Section 6 refine these requirements and trace back to them by their identifiers.

The evidence comes from three places. The Problem Statement and Goals document records the client's vision and the target users. Two potential users were interviewed on October 14, 2026, for the team's Design Thinking report; they are referred to as P01 and P02, and their statements are paraphrased or briefly quoted from de-identified transcripts. Both interviews were short and exploratory, so their answers are treated as indicative. The Hazard Analysis contributes needs that protect users from harm. Neither interviewee was a beginner at investing, so the needs of beginners come from the Problem Statement and will be checked in usability testing.

Each requirement is verifiable: it can be confirmed by inspection, demonstration, or a usability test, through the functional and non-functional requirements that refine it.

Activity and motivation

SR1: Users shall be rewarded for the physical activity they already do, without having to start, stop, or record it themselves.
Source: SH1, SH2. *Evidence:* Both participants track activity with their phone (P01, P02). Tracking should be "as touchless as possible": an app that must be started

and stopped by hand is easily left running for hours and gives wrong results (P01). Approximate step counts are acceptable to users (P01). *Goal:* G5.

SR2: Rewards for activity shall have a visible use to the user beyond the activity count itself. *Source:* SH1, SH2. *Evidence:* A long step streak gives one participant nothing but pride (P01); rewards have not changed the other's habits, although a tangible reward might make them work out more (P02). *Goal:* G5.

SR3: Users of any fitness level or physical ability shall be able to earn rewards, whatever kinds of activity they do.
Source: SH1, SH6. *Evidence:* Participants' activity goes well beyond walking: basketball, skating, and pickup hockey (P01); soccer, weightlifting, climbing stairs, and rock climbing (P02). The target users have any level of fitness (Problem Statement). *Goal:* G5.

SR4: Users shall choose their own activity goals, or set none, and shall not be pressured to raise them.
Source: SH1. *Evidence:* One participant has kept the same step goal for over a year, because a higher one takes too much time (P01); the other sets no goals and keeps only a mental check that they "at least did something" (P02). *Goal:* G5.

SR5: Features that motivate activity shall not encourage users to exercise when it is unsafe, such as late at night, while injured, or beyond a healthy amount.
Source: SH1, SH6. *Evidence:* A streak now makes one participant go out walking at 9 p.m. to reach their goal (P01); the other admires a friend for whom "it doesn't matter how much anything hurts" (P02). *Goal:* G5.

Investing practice

SR6: Users shall be able to practice investing without any risk to real money.
Source: SH1, SH2. *Evidence:* One participant was well-versed in financial math from university courses but did not know how to act on it until they had hands-on experience with real investments (P02); the other first learned by tracking their own transactions by hand (P01). *Goal:* G4.

SR7: Investing practice shall follow the real market closely enough that its lessons carry over to real investing.
Source: SH1, SH2. *Evidence:* The client's vision is a portfolio that follows real-world prices (Problem Statement). *Goal:* G4.

SR8: Users shall be able to practice in past market conditions, such as a downturn, without affecting their own progress.
Source: SH1, SH2. *Evidence:* Practicing a downturn safely prepares users for one with real money (Problem Statement). *Goal:* G4.

SR9: FitFolio shall never give, or be mistaken for, advice on what to do with real money.
Source: SH1, SH6, SH9. *Evidence:* Self-taught investors learn from YouTube and Reddit, where it is easy to follow "the wrong kinds of investing YouTubers," and look for answers tailored to their own situation (P01). Users may act on whatever the app appears to recommend. *Goal:* G4.

Learning

SR10: Users shall be able to understand each financial and fitness concept where it appears, without leaving FitFolio.
Source: SH1. *Evidence:* Beginners are told that investing requires studying measures such as price-to-earnings ratios, which confuses them (P01); advice is scattered across sources that contradict each other (Problem Statement). *Goal:* G1, G2.

SR11: Users shall learn through activities they take part in, not only by reading or watching.
Source: SH1. *Evidence:* One participant has trouble learning from videos and needs "a component that's interactive"; watching alone "doesn't stick" (P02). The other learned investing by doing it themselves (P01). *Goal:* G1.

SR12: Users shall be able to find out why their portfolio changed after their own decisions, including how diversified it is.
Source: SH1, SH2. *Evidence:* One participant considers buying individual stocks "almost like gambling" compared with diversified index investing (P01); users need to see this in their own results. *Goal:* G1.

SR13: Users shall be able to learn how the concepts they practice apply to saving and investing in Canada, such as tax-free savings accounts.
Source: SH1, SH6. *Evidence:* Both participants named Canada's account types, such as Tax-Free Savings Accounts (TFSAs) and Registered Retirement Savings Plans (RRSPs), as important things they had to learn (P01, P02); "so many people aren't doing anything at all" with their savings (P02). *Goal:* G1.

SR14: Users shall be able to see how their activity and their investments have changed over time, in one place.
Source: SH1. *Evidence:* One participant checks their heart rate often, out of curiosity and to see that it is healthy (P02); the other tracks both their steps and their portfolio, in separate tools (P01). *Goal:* G2.

Sharing and privacy

SR15: Users shall be able to share the progress they choose with friends.
Source: SH1, SH5. *Evidence:* Being ahead of friends "feels great," and good friends help those who fall behind (P02). *Goal:* G3.

SR16: Users shall control exactly what their friends can see, and nothing shall be shared without the user's choice.
Source: SH1, SH5. *Evidence:* Removing names does not make a person anonymous, because the stories they tell and other context can reveal who they are (P01). *Goal:* G3.

SR17: FitFolio shall not make users feel that they are falling far behind other users.
Source: SH1, SH6. *Evidence:* Comparison is fine when slightly behind, and "only ever bad when you're really far behind" (P02). *Goal:* G3.

SR18: Users' personal and health data shall be collected only where it is needed, kept secure, and never sold.
Source: SH1, SH9. *Evidence:* One participant avoided step counters from large companies, turned off their step counter's network access, and keeps their portfolio records on their own computer (P01); the other shares data only with a reputable company that keeps it secure (P02). Required by PIPEDA (Government of Canada, 2000). *Goal:* G3.

SR19: Users shall understand why each permission is needed before they grant it.
Source: SH1, SH7. *Evidence:* Both participants grant a permission when there is a clear reason for it (P01, P02), and distrust apps that ask for "12 permissions" when two would do (P01). *Goal:* G2, G3.

SR20: Users shall be able to withdraw their consent and delete their account and data.
Source: SH1, SH9. *Evidence:* Users' right to withdraw consent is required by PIPEDA (Government of Canada, 2000). *Goal:* G3.

Access and operation

SR21: Users shall be able to install FitFolio from their phone's app store and start earning rewards without outside help.
Source: SH1, SH2. *Evidence:* First-time users must complete the core tasks without outside help (Problem Statement Goal 2); how well an app is designed affects whether users trust it (P02). *Goal:* G2.

SR22: Users shall be able to use every core feature with only their phone, on either Android or iOS, without a wearable device.
Source: SH1, SH2. *Evidence:* One participant uses an Android phone and no longer wears a fitness band (P01); the other uses an iPhone and an Apple Watch (P02). *Goal:* G2, G5.

SR23: Users with visual or motor disabilities shall be able to use the core features.
Source: SH1, SH6, SH9. *Evidence:* Accessibility is part of serving every user and is expected by app stores (WCAG 2.2 (World Wide Web Consortium, 2023)). *Goal:* G2.

SR24: Administrators shall be able to update the tradable assets and the educational content without releasing a new version of the app.
Source: SH4. *Evidence:* Educational content must be corrected and reviewed over time (Hazard Analysis), and the list of assets will change. *Goal:* G1, G4.

SR25: FitFolio shall use health and market data only within the terms and limits set by their providers.
Source: SH7, SH8. *Evidence:* Providers limit what data can be read and how often prices can be requested (Problem Statement). *Goal:* G4, G5.

Table 3 traces each stakeholder requirement to the stakeholders it comes from. Every stakeholder in Table 1 except the development team (SH3), whose needs are expressed through the qualities of the requirements themselves, is the source of at least one requirement.

Table 3: Traceability between stakeholders and stakeholder requirements

	Stakeholder								
Requirement	**SH1**	**SH2**	**SH3**	**SH4**	**SH5**	**SH6**	**SH7**	**SH8**	**SH9**
SR1	×	×							
SR2	×	×							
SR3	×					×			
SR4	×								
SR5	×					×			
SR6	×	×							
SR7	×	×							
SR8	×	×							
SR9	×					×			×
SR10	×								
SR11	×								
SR12	×	×							
SR13	×					×			
SR14	×								
SR15	×				×				
SR16	×				×				
SR17	×					×			
SR18	×								×
SR19	×						×		
SR20	×								×
SR21	×	×							
SR22	×	×							
SR23	×					×			×
SR24				×					
SR25							×	×	

4 User Personas

The personas below represent the main kinds of user FitFolio is built for. They are fictional, but each is based on the evidence named in its *Based on* entry: Personas 1 and 2 on the two interviews (P01 and P02), and Persona 3 on the target users in the Problem Statement, since neither interviewee was a beginner at investing. Each persona combines needs from that evidence with invented details, so no persona describes a real person. Each persona lists the stakeholder requirements that matter most to them, so the design can be checked against the people it serves.

4.1 Persona 1: Sam

- **Name:** Sam, 27, works full time.
- **Description:** Walks most days and has tracked steps on an Android phone for years. A long streak at a fixed step goal sometimes gets Sam out walking late in the evening. Sam also plays recreational sports that the phone does not track, and prefers not to wear a fitness band. Sam taught themselves investing from free online content and prefers simple, diversified index investing.
- **Goals:** Keep the streak going without it taking over the evening; get credit for sports, not only steps; keep learning investing in a way that fits their own situation; keep control of their data.
- **Frustrations:** Apps that must be started and stopped by hand; investing content that pushes risky stock-picking; apps that ask for far more permissions than they need or that may sell their data.
- **Technical Ability:** High. Sam chooses apps for privacy and limits what each app can access.
- **Based on:** P01.
- **Most relevant requirements:** SR1, SR3, SR5, SR9, SR18, SR19, SR22.

4.2 Persona 2: Jordan

- **Name:** Jordan, 25, recent graduate.
- **Description:** Studied some finance at school, but only understood how to act on it after starting to invest with real money. Is active now and then, but exercise is not a priority right now. Jordan's phone and smartwatch record their activity, and they check their health data out of curiosity. Jordan sets no fitness goals and keeps only a rough sense of whether they were active that week.

- **Goals:** Be more consistent with exercise, like a friend Jordan admires; learn by doing rather than by watching; understand their account options well enough that their savings grow instead of falling behind inflation.
- **Frustrations:** Learning from videos that do not stick; falling far behind friends; apps that ask for personal data without a clear reason.
- **Technical Ability:** Moderate to high. Uses a phone and smartwatch daily, and judges whether to trust an app by who makes it and how well it is designed.
- **Based on:** P02.
- **Most relevant requirements:** SR2, SR11, SR13, SR15, SR17, SR4, SR19.

4.3 Persona 3: Aisha

- **Name:** Aisha, 19, first-year student.
- **Description:** Has never invested and does not know what a stock, an exchange-traded fund, or a tax-free savings account is. Her savings sit in a bank account. She walks between classes but does not exercise regularly. She would like to start investing, but is afraid of losing money she cannot afford to lose and has heard that investing means studying company finances.
- **Goals:** Understand the basics of investing in plain language; start a fitness habit without being judged; try things without real consequences.
- **Frustrations:** Financial jargon; advice that contradicts other advice; apps that assume she already knows how they work.
- **Technical Ability:** Everyday smartphone use: she can install an app and respond to permission prompts, but has never used a fitness or investing app.
- **Based on:** the target users described in the Problem Statement, and both interviewees' observations that many people do nothing with their savings (P02) and that beginners are put off by the belief that investing requires expert analysis (P01). This persona will be validated in usability testing.
- **Most relevant requirements:** SR6, SR10, SR13, SR21, SR3, SR14.

5 Use Cases

Each use case below describes how a user or administrator works with FitFolio to reach a goal, with its preconditions, main success scenario, alternative flows and postconditions. The functional requirements in Section 6 trace to these use cases.

5.1 UC1: Record Daily Activity

Primary Actor

User

Preconditions

- The user has granted permission to access health data.

Main Success Scenario

1. The user opens their activity summary.
2. The system retrieves the daily activity data.
3. The system displays the recorded activity.

Alternative Flows

- If permission is denied, the system informs the user.
- If no activity data is available, the system displays that no activity was recorded.

Postconditions

- The user can view their recorded daily activity.

5.2 UC2: Earn Virtual Currency

Primary Actor

User

Preconditions

- The user has recorded a supported activity.

Main Success Scenario

1. The system retrieves the user's recorded activity.
2. The system calculates the virtual currency earned.
3. The system applies the daily earning limit.
4. The user's currency balance is updated.

Alternative Flows

- If no eligible activity is recorded, no currency is awarded.
- Activity exceeding the daily cap earns no additional currency.

- If an activity record exceeds the plausibility limits for its type, the system does not count it toward currency, trends, streaks or milestones and informs the user (HA SEC-1).

Postconditions

- The user's currency balance reflects the earned rewards.

5.3 UC3: View Market Prices

Primary Actor

User

Preconditions

- The user has an Internet connection.

Main Success Scenario

1. The user opens the simulated market.
2. The user selects an asset.
3. The system displays its latest available market price.
4. The system displays when the price was obtained.

Alternative Flows

- If market data is unavailable, the system informs the user.
- If the price is outdated, the system marks it as delayed.

Postconditions

- The user can view the available price of the selected asset.

5.4 UC4: Buy a Simulated Asset

Primary Actor

User

Preconditions

- The user has enough virtual currency for the purchase.
- A usable market price is available.

Main Success Scenario

1. The user selects an asset to purchase.
2. The user enters the quantity.
3. The system displays the total cost.
4. The user confirms the transaction.
5. The system updates the balance and holdings.

Alternative Flows

- If the user has insufficient currency, the purchase is rejected.
- If the transaction fails, the balance and holdings remain unchanged.
- If the market price is outdated, the system does not allow the trade and informs the user.
- If the market is closed, the system queues the trade until the market reopens.
- If the user submits the same trade twice, the system only processes it once.
- If this is the user's first trade, they must acknowledge that the market is simulated and does not provide financial advice.
- If the user chooses to sell instead, the system checks their shares and adds the virtual currency from the sale to their balance.

Postconditions

- The purchase is recorded and the portfolio is updated.

5.5 UC5: Practice a Historical Market Scenario

Primary Actor

User

Preconditions

- The user is signed in.

Main Success Scenario

1. The user opens the market scenarios section.
2. The user selects a historical market scenario.
3. The system displays the historical period being replayed.

4. The user buys or sells simulated assets using the scenario's market prices.
5. The system updates the user's scenario balance and holdings.
6. The user views the results of their trades.

Alternative Flows

- The user can leave the scenario at any time.
- If the user does not have enough scenario currency or shares, the system does not allow the trade.

Postconditions

- The user's regular virtual currency balance and holdings remain unchanged.

5.6 UC6: Choose Reminder Times

Primary Actor

User

Preconditions

- The user can access their account settings.

Main Success Scenario

1. The user opens reminder settings.
2. The user selects preferred reminder periods.
3. The system saves the settings.

Alternative Flows

- If reminders are disabled, no optional reminders are sent.

Postconditions

- The user's reminder preferences are saved.

5.7 UC7: Withdraw Data Consent

Primary Actor

User

Preconditions

- The user previously granted data access.

Main Success Scenario

1. The user opens data permission settings.
2. The user selects a permission to withdraw.
3. The system records the change.
4. The system stops reading the selected data.

Alternative Flows

- If operating system settings are required, the user is directed there.

Postconditions

- The withdrawn data is no longer accessed.

5.8 UC8: Review Data Permissions

Primary Actor

User

Preconditions

- The user is signed in.

Main Success Scenario

1. The user opens permission settings.
2. The system displays the available permissions.
3. The system explains which features use each permission.

Alternative Flows

- Permissions that are not granted are marked as unavailable.

Postconditions

- The user understands the application's data access.

5.9 UC9: Share Progress with Friends

Primary Actor

User

Preconditions

- The user has at least one accepted friend connection.

Main Success Scenario

1. The user selects a statistic to share.
2. The user selects the friends who can view it.
3. The system publishes the selected information.

Alternative Flows

- If the user cancels, nothing is shared.
- Location and exact activity times are excluded.

Postconditions

- Only selected friends can view the shared statistic.

5.10 UC10: Delete Account

Primary Actor

User

Preconditions

- The user is signed in.

Main Success Scenario

1. The user selects account deletion.
2. The system requests confirmation.
3. The user confirms the request.
4. The system disables the account and deletes its associated data.

Alternative Flows

- If the user cancels, their account remains unchanged.

Postconditions

- The account is disabled and deletion is initiated.

5.11 UC11: Create and Manage Account

Primary Actor

User

Preconditions

- The user has an Internet connection.

Main Success Scenario

1. The user opens the application and selects account creation.
2. The user enters their account information.
3. The system creates the account.
4. The user signs in and accesses their profile.

Alternative Flows

- If the account information is invalid, the system asks the user to correct it.
- If the user forgets their password, they can request a password reset.
- The user can edit their profile, download their data, or sign out.

Postconditions

- The user's account and profile information are saved.

5.12 UC12: Learn Financial and Fitness Concepts

Primary Actor

User

Preconditions

- The user can access the learning content.

Main Success Scenario

1. The user selects a financial or fitness concept.
2. The system displays an explanation of the concept.
3. The user reads the explanation.

Alternative Flows

- The user can open an explanation from another part of the application.
- If the content is unavailable, the system informs the user.

Postconditions

- The user has viewed the selected concept explanation.

5.13 UC13: Review Investment Feedback

Primary Actor

User

Preconditions

- The user has a simulated investment portfolio.

Main Success Scenario

1. The user opens their portfolio.
2. The system displays the portfolio's current value and holdings.
3. The system displays feedback about changes in portfolio value.
4. The user reviews the feedback.

Alternative Flows

- If no significant change occurred, no change feedback is shown.
- If one holding exceeds the concentration limit, the system explains diversification.

Postconditions

- The user can review how their investments affected their portfolio.

5.14 UC14: Complete Knowledge Check

Primary Actor

User

Preconditions

- The user can access the learning section.

Main Success Scenario

1. The user selects a knowledge check.
2. The system displays questions about the selected concept.
3. The user submits their answers.
4. The system displays the results and explanations.
5. The system updates the user's learning progress.

Alternative Flows

- If the user exits before submitting, no completed result is recorded.

Postconditions

- The user's learning progress reflects the completed knowledge check.

5.15 UC15: Manage Friends and Social Feed

Primary Actor

User

Preconditions

- The user is signed in.

Main Success Scenario

1. The user opens the social section.
2. The user sends a friend request to another user.
3. The other user accepts the request.
4. The system adds the users as friends.
5. The user views updates shared by their friends.

Alternative Flows

- If the request is declined, the users are not added as friends.
- The user can remove an existing friend.
- If no updates are available, the social feed is empty.

Postconditions

- The user's friend connections and social feed reflect the available information.

5.16 UC16: Manage Educational Content

Primary Actor

Administrator

Preconditions

- The administrator has access to educational content management.

Main Success Scenario

1. The administrator opens the educational content.
2. The administrator selects a concept explanation or knowledge check.
3. The administrator adds or edits the content.
4. The system saves the changes.

Alternative Flows

- If the administrator cancels, no changes are saved.
- If saving fails, the previous content remains unchanged.

Postconditions

- The updated educational content is available to users.

5.17 UC17: Manage Goals and Streaks

Primary Actor

User

Preconditions

- The user is signed in.

Main Success Scenario

1. The user opens their activity goals.
2. The user sets or changes their daily activity goal.
3. The system saves the goal for the next day.
4. The user views their current streak and milestones.

Alternative Flows

- The user can earn currency without setting a goal.
- The user can mark a rest day without losing their streak.
- If the user misses their goal, no currency is taken away.

Postconditions

- The user's goal and streak information are updated.

6 Functional Requirements

This section lists what FitFolio shall do, grouped by feature area. Each requirement has a fit criterion and traces to the stakeholder requirements and use cases it refines and to the goals in Section 2.2. A requirement that begins with a Hazard Analysis (HA) identifier, such as (HA SEC-3), appears in the Hazard Analysis with the same identifier and wording. A requirement whose traces end with a Hazard Analysis identifier, such as Hazard Analysis SEC-13, implements part of that Hazard Analysis requirement.

6.1 Accounts and Profile

FR1: The system shall let a new user create an account with an email address and a password.
Fit criterion: A test user who enters an unused email address and a valid password has an account, and can sign in with them.
Traces to: SR21, G2

FR2: (HA SEC-37) The system shall not create an account until the user confirms that they are at least `MIN_USER_AGE` years old.
Fit criterion: A test user who does not confirm their age during sign-up has no account created.
Traces to: SR18, G3, Hazard Analysis CA2

FR3: The system shall require each user to choose a username that no other user has.
Fit criterion: A test user who tries to sign up with a username that another account already uses is asked to choose a different one.
Traces to: SR15, G3

FR4: The system shall let a user sign in with their email address and password.
Fit criterion: A test user who enters their correct email address and password is signed in; one who enters a wrong password is not.
Traces to: SR18, G2

FR5: The system shall let a signed-in user sign out.
Fit criterion: After a test user signs out, opening the app shows the sign-in screen and none of their data.
Traces to: SR18, G3

FR6: (HA SEC-40) The system shall let a user who has forgotten their password set a new one through a link sent to their account's email address.
Fit criterion: A test user who requests a reset receives a link at their email address, and after setting a new password can sign in only with the new one.
Traces to: SR20, G2

FR7: The system shall let the user view and edit their profile details.
Fit criterion: A test user who changes a profile detail sees the new value the next time they open their profile.
Traces to: SR20, G3

FR8: The system shall not require the user to enter their height or weight in order to use any feature.
Fit criterion: A test user who leaves height and weight empty can earn currency, trade, learn, and share.
Traces to: SR3, SR18, G3, G5

FR9: The system shall let the user download a copy of the personal data FitFolio stores about them.
Fit criterion: A test user who requests a copy receives a file containing their profile, activity totals, balance, holdings, trades, and shared updates.
Traces to: SR20, G3

FR10: Before deleting an account, the system shall ask the user to confirm the deletion in a message stating that it cannot be undone.
Fit criterion: A test user who starts account deletion is shown this warning, and their account still exists if they cancel.
Traces to: SR20, UC10, G3

FR11: The system shall let the user delete their account from within the app.
Fit criterion: A test user who confirms deletion can no longer sign in, and none of their data is shown to them or to their friends.
Traces to: SR20, UC10, G3, Hazard Analysis SEC-18

6.2 Activity and Rewards

A supported activity type is one listed in `SUPPORTED_ACTIVITY_TYPES`. For each supported type, FitFolio reads the user's activity records from the health platform, shows the daily totals in their activity trends, and awards currency for them. Each activity record counts under one supported type only, so walks and runs count as steps, not as workout minutes. Section 6.7 defines the daily reward and the streak.

FR12: The system shall award virtual currency for each activity type in `SUPPORTED_ACTIVITY_TYPES`.
Fit criterion: A tester records activity of each type in `SUPPORTED_ACTIVITY_TYPES` in the health platform, and each one earns currency.
Traces to: SR3, UC2, G5

FR13: The system shall award virtual currency for recorded activity of a supported activity type.
Fit criterion: For a test user whose recorded activity in one calendar day is plausible and would earn at most `MAX_DAILY_CURRENCY`, the balance increases by the daily reward R_d defined in Section 6.7.
Traces to: SR1, UC2, G5

FR14: (HA SAF-12) Before the user first earns currency, the system shall display a notice stating that FitFolio does not give medical advice and that users should consult a health professional before starting a new exercise program, and shall not award currency until the user acknowledges it.
Fit criterion: A new test user is shown the notice before their first reward, and earns no currency until they acknowledge it.
Traces to: SR5, UC2, G5

FR15: (HA SAF-13) The system shall award currency for recorded activity without requiring any interaction with FitFolio while the activity is in progress.
Fit criterion: A test user who records a workout without opening FitFolio during it receives the same currency as one who opens the app during the workout.
Traces to: SR1, UC2, G5

FR16: The system shall treat a day with no activity data from the health platform as a day with zero activity.
Fit criterion: For a test user whose health platform reports no activity data for a day, that day shows zero activity and earns no currency.
Traces to: SR1, UC1, G5, Hazard Analysis SEC-21

FR17: (HA SEC-1) The system shall not count an activity record that exceeds the plausibility limits for its activity type toward the user's currency, activity trends, streak or milestones, and shall show the user that the record was not counted.
Fit criterion: A test record of more than `MAX_DAILY_STEPS` steps or `MAX_DAILY_PUSHES` wheelchair pushes in one day, or a workout longer than `MAX_WORKOUT_DURATION`, earns no currency, does not appear in the user's activity trends, streak or milestones, and is shown to the user as not counted.
Traces to: SR1, UC2, G5

FR18: (HA SAF-7) The system shall award a user no more than `MAX_DAILY_CURRENCY` virtual currency for recorded activity in any calendar day.
Fit criterion: For a test user whose recorded activity in one calendar day is plausible and would otherwise earn more than `MAX_DAILY_CURRENCY`, the balance increases by exactly `MAX_DAILY_CURRENCY` for that day. Each activity counts toward the calendar day in the user's time zone at the time of the activity.
Traces to: SR5, UC2, G5

FR19: The system shall let the user set or change a daily activity goal for a supported activity type, with each change applying from the next day.
Fit criterion: A test user who changes their daily activity goal keeps the previous goal for the rest of that day and has the new goal from the next day.
Traces to: SR4, UC17, G5

FR20: (HA SAF-3) The system shall not require the user to set an activity goal to earn currency for recorded activity.
Fit criterion: A test user with no activity goal set who records activity receives currency for it.
Traces to: SR4, UC17, G5

FR21: (HA SAF-4) The system shall not change a user's activity goal without the user's confirmation.
Fit criterion: A test user who exceeds their goal on `GOAL_REVIEW_PERIOD` consecutive

days still has the same goal unless they confirm a change.
Traces to: SR4, UC17, G5

FR22: The system shall show the user their current streak, as defined in Section 6.7.
Fit criterion: A test user with a daily step goal and no streak meets the goal on three consecutive days and is shown a streak of 3. After a following rest day, they are shown a streak of 4. After a following day on which they miss the goal and take no rest day, they are shown a streak of 0. A test user with no goal set has their streak increase by one on each day they record any activity.
Traces to: SR14, UC17, G5

FR23: (HA SAF-1) The system shall let the user mark up to `REST_DAYS_PER_PERIOD` days in each `STREAK_PERIOD` as rest days, and a rest day shall not end the user's activity streak.
Fit criterion: A test user with a streak who marks a day as a rest day and records no activity that day does not have the streak reset to zero at the start of the following day. A request to mark more than `REST_DAYS_PER_PERIOD` rest days in one `STREAK_PERIOD` is refused.
Traces to: SR5, UC17, G5

FR24: (HA SAF-2) The system shall not reduce a user's virtual currency balance or simulated holdings when they miss an activity goal or their streak ends, and shall not show a missed goal or an ended streak to other users.
Fit criterion: For a test user who misses their goal on day *d*, the currency balance and simulated holdings at the start of day *d+1* equal those at the end of day *d*. No friend's social feed shows the missed goal or the ended streak.
Traces to: SR5, SR17, UC17, G3, G5

FR25: The system shall show the user each milestone in `MILESTONES` when they reach it.
Fit criterion: A test user whose streak, or total currency earned, first reaches a value in `MILESTONES` is shown that milestone once.
Traces to: SR14, UC17, G5

FR26: (HA SAF-14) The system shall not base any reward, goal or milestone on the user's body weight, change in weight, or calories.
Fit criterion: No reward rule, goal or milestone in FitFolio uses body weight, weight change or calories as an input, and a test user's rewards are unchanged when their weight is changed or removed.
Traces to: SR5, SR3, UC2, G5

FR27: The system shall show the user their daily totals for each supported activity type over time.
Fit criterion: A test user with recorded activity on 14 days sees the total for each of those days, for each supported activity type.
Traces to: SR14, UC17, G2, G5

6.3 Simulated Market and Portfolio

The live market is the simulated market that follows current real-world prices. A market scenario replays historical prices instead, with its own balance and holdings (FR45). Scenario prices are the replayed day's closing prices (FR47), so the rules on market hours and price age (FR33, FR39 and FR40) apply only to the live market. Every asset is priced in `QUOTE_CURRENCY`, and one unit of virtual currency buys one unit of `QUOTE_CURRENCY` worth of an asset. Section 6.7 defines trades and portfolio value.

FR28: The system shall offer at least `MIN_TRADABLE_STOCKS` stocks for trading in the live market.
Fit criterion: A tester counts the stocks that can be traded in the live market and finds at least `MIN_TRADABLE_STOCKS`. Exchange-traded funds offered in addition are not counted.
Traces to: SR7, UC3, G4

FR29: The system shall offer only assets whose prices are quoted in `QUOTE_CURRENCY`.
Fit criterion: Every tradable test asset has its price quoted in `QUOTE_CURRENCY`.
Traces to: SR7, UC3, G4, Hazard Analysis SEC-27

FR30: The system shall let an administrator add an asset to, or remove an asset from, the live market without releasing a new version of the app.
Fit criterion: After an administrator adds a test asset, users can trade it, and after the administrator removes it, users can no longer buy it, with no app update in between.
Traces to: SR24, G4

FR31: The system shall let a user who holds shares of a removed asset continue to see and sell those shares.
Fit criterion: After an administrator removes a test asset that a test user holds, the user still sees the holding and can sell it.
Traces to: SR6, SR24, G4

FR32: The system shall price each asset in the live market, and each removed asset a user still holds, at the most recent price it has obtained for that asset from the market data provider.
Fit criterion: For a test asset in the live market, the price FitFolio shows and uses for trades equals the latest price FitFolio received from the market data provider for that asset.
Traces to: SR7, UC3, G4

FR33: (HA SEC-2) The system shall display each asset price with the time it was obtained, and shall mark the price as delayed when the asset's market is open and the price is older than `QUOTE_STALENESS_LIMIT`.
Fit criterion: While a test asset's market is open, a price obtained more than `QUOTE_STALENESS_LIMIT` ago is shown with its time and a "Delayed" marker, and

a price obtained within the limit is shown with its time and no marker. While the market is closed, the last price is shown with its time and no marker.
Traces to: SR7, UC3, G4

FR34: (HA SAF-8) Before the user's first trade, the system shall display a notice stating that the market is simulated, that virtual currency and holdings have no real-world value, and that nothing in FitFolio is financial advice. The system shall not complete the first trade until the user acknowledges the notice.
Fit criterion: A new test user who attempts their first trade is shown the notice and cannot complete the trade until they acknowledge it. The notice contains all three statements: the market is simulated, currency and holdings have no real-world value, and nothing in FitFolio is financial advice.
Traces to: SR9, UC4, G4

FR35: The system shall let the user buy whole shares of an asset using virtual currency.
Fit criterion: While a test asset's market is open, a test user with enough currency buys one share of it. Their balance decreases by the price shown when they confirmed the trade, and their holding of it increases by one.
Traces to: SR6, SR11, UC4, G4

FR36: The system shall let the user sell whole shares of an asset they hold for virtual currency.
Fit criterion: While a test asset's market is open, a test user holding one share of it sells it. Their balance increases by the price shown when they confirmed the trade, and their holding of it decreases by one.
Traces to: SR6, SR11, UC4, G4

FR37: (HA SEC-5) The system shall record each trade completely, updating both the currency balance and the holdings, or not at all. It shall reject a purchase that costs more than the user's currency balance and a sale of more shares than the user holds.
Fit criterion: A test trade interrupted before it completes leaves the balance and holdings unchanged, and a completed trade changes both. A purchase costing more than the balance, or a sale of more shares than are held, is rejected and changes neither.
Traces to: SR7, UC4, G4

FR38: (HA SEC-6) The system shall record a trade only once when the same trade request is submitted more than once.
Fit criterion: When a test user submits a trade and the same request is sent again, for example by retrying after a dropped connection, exactly one trade is recorded.
Traces to: SR7, UC4, G4

FR39: (HA SEC-3) The system shall not execute a trade while the asset's market is open and the asset's price is older than `QUOTE_STALENESS_LIMIT`.
Fit criterion: While a test asset's market is open and its price is older than `QUOTE_STALENESS_LIMIT`, a trade request for it is refused and the user is told the price

is out of date. With a price within the limit, the trade proceeds.
Traces to: SR7, UC4, G4

FR40: (HA SEC-4) The system shall queue a trade requested while the asset's market is closed, and execute it at the first price obtained after the market next opens. If the user's balance or holdings cannot cover the trade at that price, the system shall reject it and tell the user.
Fit criterion: A test trade requested while the market is closed is shown to the user as queued, with a notice that it will run at the opening price, and is executed at the first price obtained after the market opens. A queued purchase that costs more than the user's balance at that price, or a queued sale of more shares than the user holds, is rejected, and the user is told.
Traces to: SR7, UC4, G4

FR41: The system shall show the user each of their holdings.
Fit criterion: A test user who owns shares of two assets sees both holdings, each with the number of shares held.
Traces to: SR14, UC13, G2, G4

FR42: The system shall show the user their portfolio value.
Fit criterion: For a test user with known holdings, the portfolio value shown equals V as defined in Section 6.7: the sum of each holding multiplied by its asset's most recent price.
Traces to: SR14, UC13, G2, G4

FR43: (HA SAF-10) The system shall label every displayed return and portfolio value as simulated.
Fit criterion: Every screen that shows a return or portfolio value shows a "Simulated" label next to it.
Traces to: SR9, UC13, G4

FR44: (HA SAF-18) The system shall show every gain or loss with a sign or a text label, not by colour alone.
Fit criterion: With the display shown in greyscale, a tester can tell every gain from every loss on every screen.
Traces to: SR23, UC13, G2, G4

FR45: The system shall let the user trade in a market scenario that replays the historical prices of a past period, such as a market downturn, without changing their own currency balance or holdings.
Fit criterion: A test user starts a scenario that replays a past market downturn, and their trades in it use the historical prices for the replayed dates. After the scenario, their currency balance and holdings outside it are the same as before it started.
Traces to: SR8, SR6, UC5, G4

FR46: The system shall start each market scenario with `SCENARIO_START_BALANCE` virtual currency and no holdings.
Fit criterion: A test user who starts a scenario has a scenario balance of `SCENARIO_START_BALANCE` and no scenario holdings.
Traces to: SR8, UC5, G4

FR47: The system shall advance a market scenario by one historical trading day each time the user chooses to continue.
Fit criterion: In a test scenario, each time the user continues, the replayed date moves to the next trading day and prices change to that day's closing prices.
Traces to: SR8, UC5, G4

FR48: (HA SAF-15) The system shall mark every screen of a market scenario as a scenario, and shall show the historical dates being replayed.
Fit criterion: Every screen shown during a test scenario carries a scenario label and the dates of the period being replayed, and no screen of the live market does.
Traces to: SR8, UC5, G4

FR49: (HA SAF-9) The system shall not recommend that a user buy or sell a specific asset.
Fit criterion: No screen or notification shown to a test user tells them to buy or sell a named asset, or presents assets as a "recommended pick".
Traces to: SR9, UC13, G1, G4

6.4 Learning and Feedback

The requirements in this subsection refer to the core concepts in Table 4. These are the concepts measured by the knowledge check in goal G1; the list is educational content that administrators can extend without changing these requirements.

Table 4: Core concepts

Domain	**Core concepts**
Investing	Stock; ETF; share; market price; portfolio value; return; diversification; risk and volatility; compound growth; market downturn and recovery
Canadian accounts	Tax-Free Savings Account (TFSA); Registered Retirement Savings Plan (RRSP)
Fitness	Each supported activity type; active minutes; rest day; streak

FR50: The system shall provide a concept explanation for each core concept in Table 4.
Fit criterion: For every core concept in Table 4, a tester can open a concept explanation of it in the app.
Traces to: SR10, SR13, G1

FR51: The system shall let the user open the concept explanation of a core concept from every screen on which that concept appears.
Fit criterion: On every screen that shows a core concept, a tester can open its concept explanation without leaving that screen.
Traces to: SR10, G1, G2

FR52: When the user's portfolio value has changed by more than `FEEDBACK_THRESHOLD` since they last viewed their portfolio, the system shall show feedback naming the holdings that contributed most to the change.
Fit criterion: For a test portfolio whose value has changed by more than the threshold, the feedback names the holding whose value changed most.
Traces to: SR12, G1

FR53: After each trade, the system shall show the user how the trade changed the share of their portfolio value held in each asset.
Fit criterion: After a test trade, the feedback shows each holding's share of portfolio value before and after the trade, and these match the values calculated from the holdings.
Traces to: SR11, SR12, G1, G4

FR54: When a single holding is more than `CONCENTRATION_LIMIT` of the user's portfolio value, the system shall show the user an explanation of diversification.
Fit criterion: A test portfolio with one holding above `CONCENTRATION_LIMIT` of its value shows the explanation, and the explanation names no asset to buy or sell (Hazard Analysis SAF-9).
Traces to: SR12, SR9, G1, Hazard Analysis SAF-9

FR55: The system shall provide a knowledge check for each core concept in Table 4.
Fit criterion: For every core concept in Table 4, a tester can start a knowledge check on it.
Traces to: SR11, G1

FR56: When the user finishes a knowledge check, the system shall show which of their answers were correct and an explanation of the correct answer to each question they got wrong.
Fit criterion: A test user who answers one question wrongly sees that question marked incorrect, together with the explanation of its correct answer.
Traces to: SR12, G1

FR57: The system shall show the user their learning progress.
Fit criterion: After a test user views a concept explanation and passes a knowledge check, both appear in their learning progress.
Traces to: SR14, G1, G2

FR58: The system shall let administrators add and edit concept explanations and knowledge checks without releasing a new version of the app.

Fit criterion: A concept explanation edited by an administrator is shown in its edited form on a test phone whose app has not been updated.
Traces to: SR24, G1

FR59: (HA SAF-16) The system shall show each concept explanation with its source and the date it was last reviewed, and each explanation and knowledge check answer shall be reviewed by a second team member before it is published.
Fit criterion: Every concept explanation shown to a test user has a source and a review date, and the content log shows a second reviewer for every explanation and answer key.
Traces to: SR24, SR10, G1

FR60: (HA SAF-17) The system shall base all feedback on a portfolio change on the user's own holdings and price changes, and shall show the data it is based on.
Fit criterion: For a test portfolio change, the feedback names the holdings and price changes it refers to, and these match the portfolio's records.
Traces to: SR12, G1

6.5 Social Feed

A user can share only the kinds of update in Table 5. Currency balances and portfolio values are not shareable, so that friends compare effort and learning rather than market results.

Table 5: Kinds of shared update

Kind	**Content**
Activity total	The user's total for one activity type over a day or a week
Streak	The user's current streak, in days. A rest day does not end a streak (Hazard Analysis SAF-1)
Fitness milestone	A fitness milestone the user has reached
Learning milestone	A knowledge check the user has passed

FR61: The system shall let the user send a friend request to another user by their username.
Fit criterion: A test user who enters another user's username sends that user a friend request, which the other user sees.
Traces to: SR15, UC15, G3

FR62: The system shall make two users friends only after the user who received the friend request accepts it.
Fit criterion: Two test users are not friends while a request between them is pending or after it is declined, and are friends once it is accepted.
Traces to: SR16, G3, Hazard Analysis SEC-36

FR63: The system shall let the user decline a friend request.
Fit criterion: After a test user declines a request, the two users are not friends and the request is no longer shown to the user who declined it.
Traces to: SR16, G3

FR64: The system shall let the user remove a friend.
Fit criterion: After a test user removes a friend, neither user sees the other's shared updates.
Traces to: SR16, G3, Hazard Analysis SEC-13 and SEC-36

FR65: The system shall let the user share an update of a kind listed in Table 5.
Fit criterion: A test user can share one update of each kind in Table 5, and cannot share anything else.
Traces to: SR15, UC9, G3

FR66: The system shall let the user choose which of their friends can see each shared update.
Fit criterion: A test update shared with one selected friend appears in that friend's social feed and in no other friend's.
Traces to: SR16, UC9, G3, Hazard Analysis SEC-13

FR67: The system shall share an update only after showing the user the update exactly as their friends will see it.
Fit criterion: For every test update, the preview shown before sharing matches what the selected friends see in their social feed.
Traces to: SR16, UC9, G3

FR68: (HA SEC-35) The system shall not include body measurements or individual activity records in any shared update.
Fit criterion: No shared update shown to a test user's friends contains the user's age, height, weight, or an individual activity record.
Traces to: SR18, G3

FR69: The system shall show the user a social feed of the updates their friends have shared with them, newest first.
Fit criterion: A test user's social feed contains every update shared with them by a friend and no others, ordered by the time each was shared.
Traces to: SR15, G3

FR70: The system shall let the user delete an update they have shared.
Fit criterion: After a test user deletes a shared update, it no longer appears in any friend's social feed.
Traces to: SR16, G3, Hazard Analysis SEC-13

FR71: (HA SAF-11) The system shall not rank users against each other by activity, currency or portfolio performance.

Fit criterion: No screen shown to a test user orders users by activity, currency or portfolio performance, or shows a user's position relative to others.
Traces to: SR17, G3

6.6 Reminders and Settings

FitFolio sends reminders of the types in Table 6, which are the same reminder types as in the Hazard Analysis.

Table 6: Reminder types

Type	Sent when	Feature area
Activity	Today is not a rest day, and either the user has not yet met today's activity goal or their streak will end if they record no activity today	Activity and Rewards
Milestone	The user meets their activity goal or reaches a value in `MILESTONES`	Activity and Rewards
Portfolio	Price changes have moved the user's simulated portfolio value by more than `FEEDBACK_THRESHOLD` since the previous day	Simulated Market and Portfolio
Learning	A new concept explanation or knowledge check is published	Learning and Feedback
Social	A friend shares an update with the user or sends them a friend request	Social Feed

FR72: The system shall let the user choose the time periods in which they receive reminders.
Fit criterion: A test user sets a reminder period, and it is shown in their settings the next time they open the app.
Traces to: SR5, UC6, G2, Hazard Analysis SAF-6

FR73: (HA SAF-6) The system shall send reminders of the types in Table 6 only during the time periods the user has chosen to receive them.
Fit criterion: For a test user who has chosen their reminder periods, no reminder of any type in Table 6 is delivered outside those periods. For a test user who has not chosen any periods, reminders are delivered only during `DEFAULT_REMINDER_PERIOD`.
Traces to: SR5, UC6, G2

FR74: The system shall let the user turn each reminder type in Table 6 on or off.
Fit criterion: A test user who turns off one reminder type receives no reminders of that type, and still receives reminders of the types left on.
Traces to: SR5, UC6, G2

FR75: The system shall send a reminder of each type that is turned on when that type's event in Table 6 occurs.

Fit criterion: For each reminder type a test user has left on, causing the event in its Sent when column produces one reminder of that type within the user's reminder periods.
Traces to: UC6, G2, G5

FR76: The system shall let the user withdraw consent for FitFolio to read their health data.
Fit criterion: After a test user withdraws consent in the app, no further health data is read.
Traces to: SR20, UC7, G3, Hazard Analysis SEC-9

FR77: The system shall show the user which health data types FitFolio has permission to read, and the feature that uses each type.
Fit criterion: A test user who has granted health permissions sees each permitted data type listed with the feature that uses it.
Traces to: SR19, SR18, UC8, G3, Hazard Analysis SEC-8

6.7 Formal Definitions

This subsection defines the daily reward, the streak, trades and portfolio value that the requirements in Sections 6.2 and 6.3 refer to. For a user on calendar day d in their time zone:

- T is the set of supported activity types (`SUPPORTED_ACTIVITY_TYPES`).
- $x_{t,d}$ is the user's total of activity type t on day d, counting only plausible records (FR17).
- u_t is the amount of activity type t that earns one unit of virtual currency (`EARNING_RATES`).
- G_d is the set of the user's activity goals in effect on day d. A goal (t, g) is met when $x_{t,d} \geq g$.
- rest_d is true when the user marked day d as a rest day.

Daily reward. The virtual currency awarded for day d is

$$R_d = \min\Big(\texttt{MAX_DAILY_CURRENCY},\ \sum_{t \in T} \Big\lfloor \frac{x_{t,d}}{u_t} \Big\rfloor\Big). \tag{1}$$

Streak. Day d counts toward the streak when

$$q_d = \mathit{rest}_d \vee \big(G_d \neq \emptyset \wedge \forall (t,g) \in G_d : x_{t,d} \geq g\big) \vee \Big(G_d = \emptyset \wedge \sum_{t \in T} x_{t,d} > 0\Big). \tag{2}$$

The streak at the end of day d is $s_d = s_{d-1} + 1$ if q_d is true, and $s_d = 0$ otherwise, with $s = 0$ before the user's first day.

Trades. Let B be the user's balance, h_a the number of shares of asset a they hold, and p_a the price at which a trade of a executes. A purchase of n shares, where n is a positive integer, is accepted only if $n \cdot p_a \leq B$, and then sets $B \leftarrow B - n \cdot p_a$ and $h_a \leftarrow h_a + n$. A sale of n shares, where n is a positive integer, is accepted only if $n \leq h_a$, and then sets $B \leftarrow B + n \cdot p_a$ and $h_a \leftarrow h_a - n$.

Portfolio value. With p_a the most recent price of asset a, the portfolio value is

$$V = \sum_a h_a \cdot p_a. \tag{3}$$

7 Data Requirements

7.1 Major Data Entities

FitFolio must manage data across user activity, fitness tracking and asset portfolio.

Table 7: Major Data Entities

Entity	Description and relationships
User Data	Represents a registered user's account and profile information, including authentication information and physical attributes such as age, height, and weight. Each user is linked to their activity records and portfolios. Also includes consent, settings, goals, rest days, and learning progress. **Location:** Both phone and server. Authentication verification data remains on the server.
Activity Record	Represents an activity belonging to a user. Contains the activity type, relevant fitness metrics such as step count and activity duration, and the amount of virtual currency received for that activity. On the server, activity is stored as daily totals per activity type, as required by SEC-10. **Location:** Both phone and server. Individual activity records and exact activity times remain on the phone.
Portfolio	Represents a user's simulated investments, including their available virtual currency, individual asset holdings, trade history, and overall portfolio performance metrics. Trade history links to the Trade entity. **Location:** Both phone and server. The server maintains the authoritative balances and holdings.

Market Data	Represents information about the assets available to trade within the application, including their names, current prices, and historical prices. This information is used to price trades and calculate portfolio values. Also records when the server obtained each price. **Location:** Both phone and server.
Trade	Represents a simulated purchase or sale belonging to a portfolio. Contains the asset, quantity, request identifier, status, and execution details. **Location:** Both phone and server. The server processes trade requests and records their outcomes.
Social Data	Represents connections between users and the shared updates they choose to show friends, of the kinds in Table 5. Records the author, kind, content and permitted viewers of each shared update. Also includes friend lists and friend requests. **Location:** Both phone and server.

7.2 Data Dictionary

Table 8: Data Dictionary

Attribute	Type	Description
User Data — Phone and server		
`user_id`	UUID	Unique identifier for the user.
`email`	String	Email address used for authentication.
`display_name`	String	Username displayed within the application.
`age_years`	Integer	Age in years.
`height_cm`	Decimal	Height in centimetres.
`weight_kg`	Decimal	Weight in kilograms.
`health_consent`	Boolean	Whether the user consents to health-data access. Withdrawing consent stops further health-data reads.
`consent_updated_at`	Timestamp	Time health-data consent was last granted or withdrawn.
`time_zone`	String	User's time zone for daily activity totals and reminders.
`reminder_settings`	List	Reminder categories, whether each is enabled, and the selected delivery windows.

goals	List	Fitness goals, including the activity type, target value, measurement unit, and effective date. Goal changes take effect the following day.
rest_days	List	Calendar dates selected as rest days for streak calculations.
learning_progress	List	Learning records containing the content identifier, completion state, knowledge-check responses, and assessment results.

Activity Record — Phone and server

activity_id	UUID	Unique identifier for the activity.
user_id	UUID	Identifier of the user who owns the activity data.
activity_type	String	Supported activity type (SUPPORTED_ACTIVITY_TYPES), such as steps or cycling.
activity_date	Date	Calendar day represented by the server's daily total. There is one total per user, date, and activity type.
start_time	Timestamp	Beginning of the activity. Stored on the phone only.
end_time	Timestamp	End of the activity. Stored on the phone only.
step_count	Integer	Number of steps recorded. On the server, this is the daily total for the activity type.
push_count	Integer	Number of wheelchair pushes recorded. On the server, this is the daily total for the activity type.
duration_seconds	Integer	Activity duration in seconds. On the server, this is the daily total for the activity type.
distance_metres	Decimal	Distance travelled in metres. On the server, this is the daily total for the activity type.
reward_amount	Decimal	Amount of virtual currency earned for the activity. On the server, this is the amount credited for the daily total, subject to the daily earning cap.

Portfolio — Phone and server

portfolio_id	UUID	Unique identifier for the portfolio.
user_id	UUID	Identifier of the user who owns the portfolio.
cash_balance	Decimal	Available virtual currency.

holdings	String	Assets owned, including each asset's identifier and share quantity.
trade_history	String	Completed trades, including purchases and sales of assets. Links to records in the Trade entity.
total_value	Decimal	Current value of all asset holdings, excluding available cash.
scenario_id	UUID	Market scenario the portfolio belongs to. Absent for the user's live portfolio, which is kept separate from every scenario portfolio.
balance_changes	List	Every change to cash_balance, with its amount, time and reason.

Market Data — Phone and server

asset_id	UUID	Unique identifier for the asset.
symbol	String	Asset's market symbol.
name	String	Asset or company name.
current_price	Decimal	Most recent available price.
price_currency	String	ISO 4217 code of the currency in which the asset's prices are quoted. Only prices in QUOTE_CURRENCY are used (SEC-27).
price_obtained_at	Timestamp	Time the server obtained the current price from the market-data provider. Reading a cached price does not change this timestamp.
price_history	String	Historical asset prices with associated timestamps.

Trade — Phone and server

trade_id	UUID	Unique identifier for the trade.
request_id	UUID	Unique identifier for the trade request. The same identifier is retained when retrying a request to prevent duplicate execution.
portfolio_id	UUID	Identifier of the portfolio placing the trade.
asset_id	UUID	Identifier of the asset purchased or sold.
trade_type	Enumeration	Purchase or sale.
quantity	Integer	Positive whole number of shares requested.
status	Enumeration	Trade state: pending, queued, executed, or rejected.
unit_price	Decimal	Price per share applied when the trade executes. Absent until execution.
currency_amount	Decimal	Virtual currency deducted for a purchase or credited for a sale. Absent until execution.

requested_at	Timestamp	Time the trade request was received.
executed_at	Timestamp	Time the simulated trade was completed. Absent until execution.

Social Data — Phone and server

user_id	UUID	Identifier of the user who owns the friend list.
friend_list	List	Identifiers of users with accepted friendships.
friend_requests	List	Friend request records containing the request identifier, sender, recipient, status, and creation time. Status is pending, accepted, or declined.
update_id	UUID	Unique identifier for the social update.
author_id	UUID	Identifier of the user who authored the social update.
created_at	Timestamp	Time the update was published.
update_kind	Enumeration	Kind of shared update, one of the kinds in Table 5.
update_content	String	Content of the shared update, as described for its kind in Table 5.
permitted_viewers	List	Identifiers of selected friends permitted to view the shared update.

7.3 Privacy and Data Retention

This subsection restates the Hazard Analysis requirements that protect the data FitFolio collects, which is listed in Section 7.2, and keep it correct. Each one keeps its Hazard Analysis identifier and wording. The other Hazard Analysis requirements appear in Section 6, except SAF-5, which applies only if goal suggestions are added, and SEC-36, which is split between FR62 and FR64.

SEC-7 (Least-privilege health access). The system shall request read access only to the health data types needed for the supported activity types (SRS Appendix C, SUPPORTED_ACTIVITY_TYPES), and shall not request permission to write to the health platform.

Fit criterion: Every health permission the system requests is a read permission for a data type needed by one of the supported activity types.

SEC-8 (State the purpose of each permission). Before requesting each permission, the system shall tell the user which feature uses that data.

Fit criterion: For every permission request, a test user is shown the feature that uses that data before the request appears.

SEC-9 (Consent and withdrawal). The system shall obtain the user's consent before it first reads their health data, and shall stop reading health data when the user withdraws consent in the app.

Fit criterion: A new test user's health data is not read until they consent. After they withdraw consent in the app, no further health data is read.

SEC-10 (Data minimization). The system shall send activity data from the user's mobile device to its server only as daily totals for each activity type, not as individual activity records.

Fit criterion: The activity data a test phone sends to the server contains only daily totals for each activity type and no individual activity records.

SEC-11 (No data to third parties). The system shall not send users' personal or health data to any third party, other than a user's email address and password reset link to the email service that delivers it. Requests to the market data provider shall identify assets only by symbol.

Fit criterion: Every outgoing request from a test mobile device and from the server goes to FitFolio's own server, the market data provider or the email service. Requests to the market data provider contain no personal or health data and identify assets only by symbol. Requests to the email service contain no personal or health data other than the user's email address and their password reset link.

SEC-12 (No location data). The system shall not read or store location or route data.

Fit criterion: The permissions a test build requests include no location permission, and no location or route data is stored on the phone or the server.

SEC-13 (Private by default). The system shall show a user's statistics only to that user until they choose to share them, and shall show a shared update only to the friends the user selected for it. When the user deletes a shared update or removes a friend, the system shall stop showing the affected shared updates to that friend.

Fit criterion: A test user's new statistic is visible to no other user. Once shared with selected friends, it is visible to exactly those friends. After the user deletes the shared update or removes one of those friends, the shared update no longer appears in that friend's social feed.

SEC-14 (No location detail in shared updates). The system shall not include location, route or the exact time an activity took place in shared updates.

Fit criterion: No shared update shown to a test user's friends contains a location, a route, or the exact time an activity took place.

SEC-15 (Authenticated access only). The system shall require an authenticated session for every request that reads or changes a user's data, and shall return another

user's data, other than their username, only when that user has shared it with the requester.

Fit criterion: A test request without a valid session is rejected. A request for another user's data, other than their username, that was not shared with the requester is rejected, and a request for data that was shared succeeds.

SEC-16 (Protect data in transit and at rest). The system shall encrypt user data in transit between the phone and the server, and when stored on the server, including in backups.

Fit criterion: Inspecting the network traffic between a test phone and the server, the data stored on the server, and its backups shows no readable user data.

SEC-17 (Session expiry). The system shall end a session when the user signs out of it or when it has not been used for `SESSION_TIMEOUT`, after which the user must sign in again.

Fit criterion: A test session is rejected after the user signs out of it, and after it is left unused for longer than `SESSION_TIMEOUT`. In both cases the user must sign in again to continue.

SEC-18 (In-app account deletion). The system shall let the user delete their account from within the app, and shall delete all data it stores about them, as listed in the SRS data dictionary (SRS Section 7.2), from the server and its backups within `DATA_DELETION_PERIOD` of the request.

Fit criterion: After a test user deletes their account in the app, they can no longer sign in. Within `DATA_DELETION_PERIOD`, neither the server nor its backups hold any data about them listed in the SRS data dictionary, including their account details, profile, activity totals, goals, rest days, reminder settings, consent records, currency balance, holdings, trades, scenario portfolios, learning progress, shared updates, friend list and friend requests, and none of their shared updates appear in friends' social feeds.

SEC-19 (Count each activity once). The system shall award currency for each activity record at most once, even when the record is read more than once.

Fit criterion: When the same test records are read on two consecutive syncs, the user is rewarded for them once.

SEC-20 (Catch late activity). On each sync, the system shall read the activity records of the last `SYNC_LOOKBACK_PERIOD` and reward any record that has not already been rewarded.

Fit criterion: A test record written to the health platform two days after the activity took place is rewarded on the next sync, once.

SEC-21 (Activity unavailable). When the system cannot read activity data, because permission has not been granted or the health platform does not respond, it shall tell

the user that activity could not be read, keep the currency already earned, and keep every feature that does not need new activity data usable.

Fit criterion: With health permission denied, and separately with the health platform returning errors, a test user is told that activity could not be read, their balance is unchanged, and trading, learning and the social feed still work.

SEC-22 (Balance history). The system shall record every change to a user's currency balance with its amount, time and reason, and the balance shall always equal the sum of its recorded changes.

Fit criterion: For a test user after a series of rewards and trades, the balance equals the sum of the recorded changes, and each change shows its amount, time and reason.

SEC-23 (Count activity on the user's day). The system shall assign each activity record to a calendar day using the user's time zone at the time of the activity.

Fit criterion: Test records at 23:30 and 00:30 local time are assigned to two different days, and a record made before a time zone change keeps the day on which it took place.

SEC-24 (The server decides rewards and balances). The system shall calculate rewards, balances and holdings only on the server, from the daily activity totals the app sends, and shall reject any reward, balance or holding reported by the app and any daily total dated after the server's current date in the user's time zone.

Fit criterion: A test request from a modified app that reports its own reward, balance or holding is rejected, and a test daily total dated after the server's current date in the user's time zone is rejected and earns nothing.

SEC-25 (Validate every request). The server shall reject any request whose inputs are missing, of the wrong type, or outside the limits stated in the SRS.

Fit criterion: Test requests with a missing field, a value of the wrong type, and a value outside its SRS limit, such as a negative trade quantity, are each rejected with an error and change no data.

SEC-26 (Backups). The system shall back up all persistent data at least once every `BACKUP_INTERVAL`, and restoring from a backup shall be tested before each release.

Fit criterion: The backup log shows a backup in every `BACKUP_INTERVAL`, and the release checklist shows a successful test restore before each release.

SEC-27 (Prices in one currency). The system shall store every price with its currency, and shall not use a price that is not in `QUOTE_CURRENCY`.

Fit criterion: Every stored test price has a currency. A test price received in a currency other than `QUOTE_CURRENCY` is not shown or used in any balance, trade or portfolio value.

SEC-28 (Apply stock splits). When an asset undergoes a stock split, the system shall adjust users' holdings and the asset's price history so that the split does not change any portfolio value.

Fit criterion: For a test asset that splits 2-for-1, every holding doubles, every historical price halves, and every portfolio value is the same before and after.

SEC-29 (Trade at the confirmed price). The system shall execute a trade made while the asset's market is open at the price shown when the user confirmed it. If that price is no longer the latest price, it shall show the user the new price and ask them to confirm again.

Fit criterion: While a test asset's market is open, a trade confirmed at a price that has not changed executes at that price. When the price changes between viewing and confirming, the trade does not execute until the user confirms the new price.

SEC-30 (Accurate values). The system shall calculate every displayed balance and portfolio value to within `VALUE_TOLERANCE` of its exact value.

Fit criterion: For test portfolios with hand-calculated values, every displayed balance and portfolio value differs from the hand-calculated value by at most `VALUE_TOLERANCE`.

SEC-31 (Separate scenario portfolios). The system shall keep each market scenario's portfolio and balance separate from the user's live portfolio and balance.

Fit criterion: After a test user buys and sells in a scenario, their live balance and holdings are unchanged.

SEC-32 (Complete scenario data). The system shall offer a market scenario only after its historical data has been checked to contain a price for every asset in it on every trading day of its period.

Fit criterion: The check is run on every scenario dataset before release, and a test dataset with one missing day is not offered.

SEC-33 (Provider credentials stay on the server). The system shall store the credentials for every external service only on the server, never in the app or in the code repository.

Fit criterion: Inspection of the app package and an automated scan of the repository find no external service credentials.

SEC-34 (Private notifications). The system shall not include health data, balances or portfolio values in a notification unless the user has turned this on.

Fit criterion: With default settings, no notification sent to a test phone contains health data, a balance or a portfolio value.

SEC-38 (Limit sign-in attempts). The system shall require passwords of at least `MIN_PASSWORD_LENGTH` characters, and after `MAX_FAILED_SIGNINS` failed sign-in attempts on an account shall refuse further attempts on it for `SIGNIN_LOCKOUT_PERIOD`.

Fit criterion: A test password shorter than `MIN_PASSWORD_LENGTH` is refused at sign-up, and after `MAX_FAILED_SIGNINS` wrong passwords even the correct password is refused until `SIGNIN_LOCKOUT_PERIOD` has passed.

SEC-39 (Protect the session on the phone). The system shall store the user's session only in the phone's secure storage.

Fit criterion: Inspection of a test phone's app storage finds the session only in the operating system's secure storage.

8 Non-Functional Requirements

NFR1: **Accuracy**

The system shall calculate virtual currency balances and simulated portfolio values within `VALUE_TOLERANCE` of their expected values.

Fit criterion: Independently calculated test cases differ from the displayed result by no more than `VALUE_TOLERANCE`.

Traces to: G4, G5, FR13, FR35, FR53, Hazard Analysis SEC-30.

NFR2: **Usability**

At least `MIN_TASK_SUCCESS_RATE` of first-time users shall be able to complete the core activity and simulated investing tasks without outside assistance.

Fit criterion: At least `MIN_TASK_SUCCESS_RATE` of usability test participants complete the core tasks independently.

Traces to: G1, G2, FR13, FR35, FR55.

NFR3: **Maintainability**

The effort required to implement a likely change shall not exceed `MAX_CHANGE_EFFORT` of the original development effort for the affected feature.

Fit criterion: A modification requires no more than `MAX_CHANGE_EFFORT` of the original feature's development hours.

Traces to: G1, G4, G5, FR58.

NFR4: **Portability**

The core features of FitFolio shall operate on supported Android and iOS devices.

Fit criterion: Core acceptance tests pass on both supported operating systems.

Traces to: G2, G3, G4, G5, SC1, SC2.

NFR5: **Performance**

The system shall display previously loaded activity summaries, portfolio summaries and social feed updates within `MAX_RESPONSE_TIME` of the user's request in at least `MIN_RESPONSE_RATE` of requests.

Fit criterion: At least `MIN_RESPONSE_RATE` of `RESPONSE_TRIALS` timed trials complete within `MAX_RESPONSE_TIME` under a stable network connection.

Traces to: G2, FR27, FR42, FR69.

NFR6: **Reliability**

The system shall preserve currency balances and portfolio holdings when trades are interrupted or repeated.

Fit criterion: In `RELIABILITY_TRIALS` interruption and retry tests, no duplicate or incomplete trades are recorded.

Traces to: G4, FR37, FR38, Hazard Analysis SEC-5 and SEC-6.

NFR7: **Security**

The system shall prevent unauthorized access to private user information.

Fit criterion: All tested requests without valid authentication or authorization are rejected.

Traces to: G3, FR4, FR5, FR66, Hazard Analysis SEC-15 and SEC-17.

NFR8: **Accessibility**

The core features shall remain usable when the device text size is increased to `MAX_TEXT_SCALE` of its default size.

Fit criterion: Users can read essential information and complete core tasks at `MAX_TEXT_SCALE` text scaling without clipped controls.

Traces to: G2, FR1, FR35, FR55, FR65.

9 AI and Machine Learning Requirements

This section is excluded because FitFolio does not use a machine learning model or any other artificial intelligence component. Rewards, plausibility checks, streaks, feedback and reminders all follow the fixed rules in Section 6, so there are no model inputs, model outputs, evaluation metrics or known model limitations to specify.

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR8, SR6	UC5	FR45
SR8	UC5	FR46
SR8	UC5	FR47
SR8	UC5	FR48
SR21	UC11	FR1
SR18	UC11	FR2
SR15	UC11	FR3
SR18	UC11	FR4
SR18	UC11	FR5
SR20	UC11	FR6
SR20	UC11	FR7
SR3, SR18	UC11	FR8
SR20	UC11	FR9
SR20	UC10	FR10
SR20	UC10	FR11
SR3	UC2	FR12
SR1	UC2	FR13
SR5	UC2	FR14
SR1	UC2	FR15
SR1	UC1	FR16
SR1	UC2	FR17
SR5	UC2	FR18
SR4	UC17	FR19
SR4	UC17	FR20
SR4	UC17	FR21
SR14	UC17	FR22
SR5	UC17	FR23
SR5, SR17	UC17	FR24
SR14	UC17	FR25
SR5, SR3	UC2	FR26
SR14	UC17	FR27
SR7	UC3	FR28
SR7	UC3	FR29
SR24	–	FR30
SR6, SR24	–	FR31
SR7	UC3	FR32
SR7	UC3	FR33
SR9	UC4	FR34
SR6, SR11	UC4	FR35
SR6, SR11	UC4	FR36

SR7	UC4	FR37
SR7	UC4	FR38
SR7	UC4	FR39
SR7	UC4	FR40
SR14	UC13	FR41
SR14	UC13	FR42
SR9	UC13	FR43
SR23	UC13	FR44
SR9	UC13	FR49
SR10, SR13	UC12	FR50
SR10	UC12	FR51
SR12	UC13	FR52
SR11, SR12	UC13	FR53
SR12, SR9	UC13	FR54
SR11	UC14	FR55
SR12	UC14	FR56
SR14	UC14	FR57
SR24	UC16	FR58
SR24, SR10	UC16	FR59
SR12	UC13	FR60
SR15	UC15	FR61
SR16	UC15	FR62
SR16	UC15	FR63
SR16	UC15	FR64
SR15	UC9	FR65
SR16	UC9	FR66
SR16	UC9	FR67
SR18	UC9	FR68
SR15	UC15	FR69
SR16	UC15	FR70
SR17	UC15	FR71
SR5	UC6	FR72
SR5	UC6	FR73
SR5	UC6	FR74
–	UC6	FR75
SR20	UC7	FR76
SR19, SR18	UC8	FR77

Table 9: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR1	NFR2, NFR7, NFR8
FR2	NFR7
FR3	NFR7
FR4	NFR2, NFR7
FR5	NFR2, NFR7
FR6	NFR7
FR7	NFR7
FR8	NFR7
FR9	NFR7
FR10	NFR2, NFR7
FR11	NFR2, NFR7
FR12	NFR1
FR13	NFR1, NFR2
FR14	NFR2
FR15	NFR2
FR16	NFR2
FR17	NFR1
FR18	NFR1
FR19	NFR2
FR20	NFR2
FR21	NFR2
FR22	NFR1
FR23	NFR2
FR24	NFR2
FR25	NFR2
FR26	NFR1
FR27	NFR1, NFR5
FR28	NFR2
FR29	NFR1
FR30	NFR3
FR31	NFR6
FR32	NFR1, NFR6
FR33	NFR2
FR34	NFR2
FR35	NFR1, NFR6, NFR2, NFR8
FR36	NFR1, NFR6
FR37	NFR1, NFR6
FR38	NFR1, NFR6

FR39	NFR1, NFR6
FR40	NFR1, NFR6
FR41	NFR2
FR42	NFR1, NFR6, NFR5
FR43	NFR2
FR44	NFR8
FR45	NFR1
FR46	NFR1
FR47	NFR1
FR48	NFR2
FR49	NFR2
FR50	NFR2, NFR8
FR51	NFR2, NFR8
FR52	NFR1
FR53	NFR1
FR54	NFR1
FR55	NFR2, NFR8
FR56	NFR2, NFR8
FR57	NFR2, NFR8
FR58	NFR3
FR59	NFR3
FR60	NFR1
FR61	NFR2
FR62	NFR7
FR63	NFR7
FR64	NFR7
FR65	NFR7, NFR8
FR66	NFR7
FR67	NFR7
FR68	NFR7
FR69	NFR2, NFR5
FR70	NFR7
FR71	NFR2
FR72	NFR2
FR73	NFR2
FR74	NFR2
FR75	NFR2
FR76	NFR7
FR77	NFR7

Table 10: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

This section explains the reasoning behind the major requirements and decisions made for FitFolio.

11.1 Simulated Investing

FitFolio uses virtual currency instead of real money so users can learn about investing without the financial risk associated with investing. This allows beginners to experiment with different investment decisions without losing any real savings. The restriction on real money transactions also keeps the system focused on education rather than financial services (SC5, FR35).

11.2 Activity Based Rewards

The system rewards users for everyday physical activity to encourage consistent exercise habits. Rather than requiring the user to do specific individual workouts, the app uses activity that is already recorded by the user's phone health platform. A daily earning limit is enforced to discourage excessive exercise for virtual rewards (SC2, FR13, FR18).

11.3 Goals and Streaks

Users can set their own activity goals and track their streaks to stay motivated. Goals are optional because not every user wants to follow a specific activity target. Rest days are also included so users do not feel pressured to exercise every day to maintain their streak. Missing a goal does not reduce the user's currency or holdings (FR19, FR20, FR23, FR24).

11.4 Financial Education

Educational explanations and knowledge checks are included because simulated trading alone may not help beginners understand their investment decisions. By providing explanations alongside market activity, users can learn financial concepts while practising them. Portfolio feedback also helps users understand the effects of their trades (FR51, FR53, FR55).

11.5 Real-World Market Data

The simulated market uses real-world prices to give users a more realistic investing experience. However, outdated prices could create unrealistic trading results. The system therefore prevents trades using stale prices while the market is open and queues trades requested while the market is closed (SC4, FR32, FR39, FR40).

Trades must also be recorded correctly to make sure users do not lose virtual currency or receive incorrect holdings. The system prevents duplicate trades and ensures that both the user's balance and holdings are updated together (FR37, FR38).

11.6 Privacy and Social Features

The social features are intended to motivate users without requiring them to share private information. Users choose which friends can view their updates, and personal body measurements are excluded from shared content. Users can also remove friends or delete shared updates to maintain control over their information (FR66, FR68, FR64, FR70, NFR7).

The system also limits the health information it collects and does not read or store location data. This helps protect users from having sensitive personal information exposed. Health data is only accessed with consent, and users can withdraw that consent when they choose (FR76, Hazard Analysis SEC-9 and SEC-12).

11.7 Security and Data Protection

Since the system stores personal information, activity totals, and simulated investment records, protecting this data is important. Users must be signed in before accessing private information, and their data must be protected during storage and transmission. Sessions also expire to reduce the risk of unauthorized access (NFR7, Hazard Analysis SEC-15 to SEC-17).

11.8 Usability and Accessibility

Because FitFolio is intended for beginners, users should not need previous investing knowledge, fitness experience or specialized fitness equipment. The main features are designed to be easy to understand, and they support different levels of physical activity and the phone's accessibility settings (NFR2, NFR4, NFR8).

12 Likely Changes

The following changes are identified in the project documents. They describe parts of FitFolio that may change as the team tests its planned integrations or adds features already listed as stretch goals.

LC1: **Activity-data updates**

The timing of FitFolio's activity-data updates may change because of restrictions imposed by the phone's health platform. The Development Plan identifies updating activity data when the user opens the app as an alternative if background updates cannot work reliably. This would affect when activity data is retrieved in UC1 (Record Daily Activity) and becomes available for awarding currency under FR13. Any change must still follow the health-platform access rules in SC3.

LC2: **Wearable device support**

FitFolio may be extended to support wearable devices, such as the Apple Watch and other compatible fitness trackers. Wearable integration is already listed as a stretch

goal in the Problem Statement and Goals. Adding this support could provide additional activity data, including workout duration and calories burned. This would affect the Activity Record entity in Table 7 and its fields in Table 8. Direct access to a wearable, rather than reading its data through the phone's health platform, would also require revising SC2. Wearables would remain optional, as required by SR22.

LC3: **Market-data access**

The market-data provider, price-update frequency, and selection of supported assets may change if the chosen service's request limits or costs prevent the planned use. The Development Plan identifies switching providers, reducing the update frequency, and limiting the asset list as possible responses. These changes would affect the provider choice in SC4, the prices obtained under FR32, and the available assets represented by the Market Data entity in Table 7. Changes to the update frequency would also need to account for the price-age display in FR33 and the restriction on trading with stale prices in FR39. Any reduction below the price-freshness or asset-coverage targets in Section 2.2 would require an agreed revision to those targets and the affected requirements.

LC4: **Historical market scenarios**

The number and selection of historical market scenarios may change depending on the availability and cost of historical price data. The Development Plan identifies offering fewer scenarios or using a selected set of stored historical datasets as possible responses if the required data is difficult to obtain. This would affect the past market conditions available under SR8 and the historical prices represented by the Market Data entity in Table 7 and its `price_history` field in Table 8.

LC5: **Social challenges and leaderboards**

FitFolio may be extended to include fitness challenges and opt-in leaderboards. Social challenges and leaderboards are listed as a stretch goal in the Problem Statement and Goals, and the Hazard Analysis identifies opt-in leaderboards as a possible future feature. The current version prohibits ranking users by activity, currency, or portfolio performance under FR71 and Hazard Analysis SAF-11. Adding leaderboards would therefore require an agreed revision to those requirements and a review of the risk of harmful comparison identified in SR17. Adding comparisons of portfolio statistics would also require revisiting the allowed updates in FR65 and Table 5. Users would still choose whether to participate and which information to share, consistent with SR16 and FR66.

13 Unlikely Changes

The following parts of FitFolio are expected to stay the same because they support the project's purpose and goals.

ULC1: **Virtual currency and simulated investing**

FitFolio is expected to continue using virtual currency with no real-world monetary value. Adding real-money trading or allowing users to exchange earned currency for cash is unlikely. The Problem Statement and Goals describes an app where beginners can practise investing without risking real money or needing to link bank or brokerage accounts. Moving to real money would change this purpose.

ULC2: **Earning rewards through physical activity**

Physical activity is expected to remain the way users earn virtual currency in FitFolio. Removing activity-based rewards is unlikely because earning currency through exercise and using it in the simulated market is the project's central idea. This connection supports the goal of helping users build fitness habits while learning about investing.

ULC3: **User control over social sharing**

Users are expected to keep control over what they share and with which friends. Removing this control is unlikely because the Problem Statement and Goals includes motivating users through friends while protecting their privacy. Only the kinds of update in Table 5 can be shared, and raw health data and portfolio values are never shared (FR65, FR68). Any changes to the social features should preserve this.

14 Appendices

The appendices give the development plan, the values of the symbolic constants, the use of generative AI, and the team's reflections.

14.1 Appendix A: Additional Materials

There are no additional materials for this version of the SRS.

14.2 Appendix B: Development Plan

The Development Plan gives the project schedule and the order in which work is done. The Hazard Analysis Roadmap states which safety and security requirements are implemented in the capstone and which are left for future work.

14.3 Appendix C: Values of Auxiliary Constants

Table 11 gives the value of each symbolic constant used in this document and in the Hazard Analysis (HA).

Constant	Value	Used in
MIN_USER_AGE	18	FR2, HA SEC-37
MIN_ANDROID_VERSION	17	SC1
MIN_IOS_VERSION	27	SC1
MARKET_DATA_PROVIDER	Finnhub (Development Plan)	SC4
SUPPORTED_ACTIVITY_TYPES	Steps (walking and running), wheelchair pushes, cycling, swimming, and other workouts such as strength training or team sports (5 types, G5)	FR12, FR13, HA SEC-7
EARNING_RATES	1 per 100 steps, 1 per 100 wheelchair pushes, and 1 per minute of cycling, swimming or another workout	FR13
MAX_DAILY_CURRENCY	300	FR13, FR18, HA SAF-7
MIN_TRADABLE_STOCKS	50 (G4)	FR28
QUOTE_CURRENCY	US dollars (USD)	FR29, HA SEC-27

Constant	Value	Used in
SCENARIO_START_BALANCE	10,000	FR46
QUOTE_STALENESS_LIMIT	15 minutes (G4)	FR33, FR39, HA SEC-2, SEC-3
FEEDBACK_THRESHOLD	5%	FR52, FR75
CONCENTRATION_LIMIT	50%	FR54
DEFAULT_REMINDER_PERIOD	9:00 to 20:00 in the user's time zone	FR73, HA SAF-6
REST_DAYS_PER_PERIOD	2	FR23, HA SAF-1
STREAK_PERIOD	calendar week (Monday to Sunday) in the user's time zone	FR23, HA SAF-1
GOAL_REVIEW_PERIOD	7	FR21, HA SAF-4
MAX_GOAL_INCREASE	10%	HA SAF-5
MILESTONES	Streaks of 7, 30 and 100 days, and 1,000, 10,000 and 100,000 currency earned in total	FR25, FR75
MAX_DAILY_STEPS	150,000	FR17, HA SEC-1
MAX_DAILY_PUSHES	100,000	FR17, HA SEC-1
MAX_WORKOUT_DURATION	12 hours	FR17, HA SEC-1
SESSION_TIMEOUT	14 days	HA SEC-17
DATA_DELETION_PERIOD	30 days	HA SEC-18
SYNC_LOOKBACK_PERIOD	7 days	HA SEC-20
BACKUP_INTERVAL	24 hours	HA SEC-26
VALUE_TOLERANCE	0.01 currency units	NFR1, HA SEC-30
MIN_TASK_SUCCESS_RATE	80%	NFR2
MAX_CHANGE_EFFORT	25%	NFR3
MAX_RESPONSE_TIME	3 seconds	NFR5
MIN_RESPONSE_RATE	95%	NFR5
RESPONSE_TRIALS	30	NFR5
RELIABILITY_TRIALS	20	NFR6
MAX_TEXT_SCALE	200%	NFR8
MIN_PASSWORD_LENGTH	8	HA SEC-38
MAX_FAILED_SIGNINS	5	HA SEC-38
SIGNIN_LOCKOUT_PERIOD	15 minutes	HA SEC-38

Table 11: Values of auxiliary constants

14.4 Appendix D: Generative AI Disclosure

This section summarizes the team's reported AI use while preparing the Software Requirements Specification (SRS). It identifies the tools used, the sections affected, and how the AI-assisted material was checked.

AI Tools Used:

The following AI tools and models were used while preparing this document:

- ChatGPT (OpenAI): GPT-6.
- Codex (OpenAI): GPT-6.1 Sol.
- Claude (Anthropic): Claude Opus 5.5.

How and Where Used:

ChatGPT was used to improve grammar and wording, help format tables and figures in LaTeX, and identify errors, typos, and inconsistencies. This assistance covered Sections 1.1-1.5 (Introduction), Section 3 (Stakeholder Requirements), Section 4 (User Personas), Section 6.1 (Accounts and Profile), Section 6.4 (Learning and Feedback), Section 6.5 (Social Feed), and constants added to Appendix C. ChatGPT also helped fix the LaTeX cross-references in Section 12 (Likely Changes).

ChatGPT also helped improve grammar and format tables, figures, and itemized lists in Section 5 (Use Cases), Section 7.3 (Privacy and Data Retention), Section 8 (Non-Functional Requirements), and Section 10 (Requirements Traceability Matrices).

Codex was used to improve wording, check grammar, and help format tables in LaTeX. This assistance covered Section 1.4 (Definitions and Acronyms), Section 7.1 (Major Data Entities), and Section 7.2 (Data Dictionary).

Claude was used to improve wording, check grammar, and help format tables and references in LaTeX for contributions to Sections 2 and 6. It also helped compare these sections with other parts of the document to identify gaps and improve consistency.

Verification of AI-Generated Content:

Team members manually reviewed the AI-assisted changes to their sections. For wording and grammar changes, they reread the revised text to make sure it preserved their original ideas and intended meaning.

For the ChatGPT-assisted work on tables and figures, the content was checked against the information provided to the tool. Team members also reviewed the formatting and checked for remaining errors and inconsistencies.

For the sections where Codex and Claude were used, the authors reread the revised sections to check that the changes still reflected their original ideas and intended meaning.

The team remains responsible for the accuracy, completeness, consistency, and appropriateness of the SRS, including any material prepared with AI assistance.

14.5 Appendix E: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

Team: What went well was grounding the requirements in evidence instead of in our own assumptions. Before writing, we interviewed two potential users, and every stakeholder requirement in Section 3 records its source stakeholder and the evidence behind it, so a reader can see why each one exists. Splitting the document through GitHub issues also worked: each section had one owner, and shared conventions, such as one format for every functional requirement, mnemonic labels, and a single appendix for symbolic constants, kept five people's writing consistent. Reviewing each other's pull requests caught real problems before submission, such as a shared table that contradicted a Hazard Analysis requirement and two different words used for the same idea. Finally, because every requirement has a label, traceability became a by-product of writing: each functional requirement names the stakeholder requirement, use case and goal it comes from, which made the traceability matrices straightforward to build.

2. What pain points did you experience during this deliverable, and how did you resolve them?

Farid: My main pain point was keeping my sections consistent with work that was changing in other branches at the same time. I wrote the Introduction, Stakeholder Requirements, User Personas and three of the functional requirement subsections, and my requirements traced to labels and Hazard Analysis IDs that lived in my teammates' branches. When Davis renumbered the Hazard Analysis requirements, four of my traces pointed to the wrong requirement without any error, and I only noticed by checking every trace against his latest list. I also caused a problem myself: my editor's auto-formatter re-indented hundreds of template lines in sections I did not own, which turned a small change into merge conflicts with three other branches. I resolved this by turning format-on-save off, restoring the original spacing, branching my functional requirements from Davis's branch instead of my own, and

making small commits that each touched only my sections. I also defined every domain term in Section 1.4 before it is used, because in our Problem Statement some words, such as "compounding", were clear to us but not to a reader outside the team. Next time I will merge `main` into my branch every day instead of once at the end, link each branch to its issue, and agree on shared identifiers before writing anything that refers to them.

Angad: One of the main challenges I faced during this deliverable was keeping my sections consistent with the requirements written by other team members. I was responsible for the use cases, non functional requirements, traceability matrices, privacy and data retention, and rationale. Since these sections depended heavily on the functional requirements, changes made by other members meant I had to go back and update my work. For example, additional functional requirements were introduced later in the deliverable, which meant I had to update the use cases and both traceability matrices to make sure everything was properly connected and included. Another challenge I faced was making sure that the privacy and security requirements matched our Hazard Analysis, especially when some requirements got renumbered or had their wording changed. I resolved these issues by reviewing the latest versions of my teammates' work, checking the references between sections, and addressing feedback on my pull request. This helped me understand how closely connected the different sections of the SRS are. For our next deliverable, I think we should communicate requirement changes earlier and review each other's work throughout the process rather than waiting until most of our sections are finished. This would help reduce the amount of rework needed near the deadline.

Davis: The main pain point for me was how many design decisions the SRS involved, and how many of them crossed between sections owned by different people. Farid and I split the functional requirements, Angad wrote the use cases and traceability matrices, and Preston wrote the data dictionary, but all of these depend on the same choices, such as what a user can share with friends and whether activity data leaves the phone as individual records or daily totals. Because we each worked in our own branch and only compared our work near the deadline, we found these mismatches late. Some had a ripple effect, for example, we had used different words for the same thing, such as "post" and "shared update", one of the shared tables contradicted a Hazard Analysis requirement on how rest days affect a streak, and no one had filled in the appendix of constants that many of our requirements relied on. As a result, we spent much of the last few days aligning our sections and resolving merge conflicts instead of improving the content. I resolved this by reviewing my teammates' open pull requests against my sections, adopting the terms Farid defined in Section 1.4, filling in the appendix of constants, and listing what I still needed from each person in my pull request. Next time, I want to communicate with the team earlier and more often. That means agreeing on shared identifiers, terms and the big design decisions at the start of a deliverable, recording each decision in the related GitHub issue as soon as it is made, and merging smaller pieces of work sooner, so that conflicts show up while they are still cheaper to fix.

Zaid: What went well was reading the project documents before writing the Likely and

Unlikely Changes. I reviewed the Problem Statement and Goals, Development Plan, SRS template, rubric, and relevant lecture material to understand what these sections needed to explain. This helped me base the changes on decisions the team had already discussed, such as wearable support and the risks around activity updates and market-data access.

At first, I was unsure whether I needed to suggest new changes or use possibilities already identified in our documents. Reviewing the instructions helped me understand that I needed to explain what could change, why, and which parts of the system would be affected. For the unlikely changes, I focused on decisions that support FitFolio's main purpose, such as virtual currency and activity-based rewards.

Another difficulty was linking my section to requirements that teammates were still writing. I checked their pull requests for the correct labels and found that some references would remain unresolved until our sections were combined.

That review also helped me catch a conflict: our stretch goals included leaderboards, but the newer safety requirements prohibited rankings. I revised the likely change to explain that future opt-in leaderboards would require reviewing the safety concerns and changing the current requirements. This showed me that checking earlier documents is only part of the work, I also need to check my section against the team's latest decisions.

Preston: A pain point we experienced during the deliverable was being unable to complete some sections without information from other team members' sections. We resolved this by communicating with each other and sharing updates so our sections remained consistent.

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

Team: Most of them. Of our 25 stakeholder requirements, 19 come directly from our two user interviews (P01 and P02), for example earning from the activity users already do, seeing progress together, controlling exactly what friends can see, and never being compared in a way that discourages someone who falls behind. Three more come from our supervisor's project vision and the Problem Statement: trading at real-world prices, practising on past market periods, and respecting the market data provider's limits. The remaining three come from regulation and standards rather than conversation: control over personal data (PIPEDA), accessibility (WCAG 2.2), and keeping educational content up to date. Because each functional requirement traces to a stakeholder requirement, the interviews indirectly shaped most of the roughly 70 functional requirements as well. The interviews also changed requirements we had already planned: one participant described a step streak that pushed them to walk late at night, which led directly to the rest-day and reminder-period requirements, and both participants preferred learning by doing, which shaped Section 6.4.

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project?

Team:

- **SFWRENG 3RA3 (Software Requirements and Security Considerations):** the basis of this SRS, including stakeholder analysis, fit criteria and traceability, and of our privacy and security requirements.
- **SFWRENG 2AA4 and 3A04 (Software Design):** modular design and information hiding for the upcoming design documents, so that likely changes, such as switching market data providers, stay local to one module.
- **SFWRENG 3S03 (Software Testing):** turning our fit criteria into test cases for the V&V Plan, and unit testing with Jest and pytest.
- **SFWRENG 3DB3 (Databases):** designing the PostgreSQL schema, and using transactions so that every trade is recorded completely or not at all.
- **SFWRENG 4HC3 (Human Computer Interfaces):** usability testing against our goal that most first-time users complete core tasks without help.
- **SFWRENG 4C03 (Computer Networks and Security):** encrypting data in transit and securing the API that the app communicates with.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project?

Team:

- **Mobile development with React Native and TypeScript** (Angad, Zaid): building one interface that runs on both Android and iOS.
- **Health platform integration** (Preston, Zaid): the permission models and data formats of Apple HealthKit and Google Health Connect, and the bridge libraries that connect them to React Native.
- **Backend development with FastAPI and PostgreSQL** (Farid, Davis): authentication, request validation, and transactions that keep balances and holdings consistent.
- **Market data and investing domain knowledge** (Davis, Farid): working within the provider's request limits, market hours, stock splits, currencies, and explaining investing concepts correctly to beginners.
- **Privacy and mobile security** (whole team): meeting PIPEDA and app store rules for health data, and securing sessions and stored data.
- **Usability testing** (whole team): planning sessions with beginners and measuring whether the app teaches what it claims to.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Team:

- **React Native:** (a) the official React Native and Expo documentation and tutorials, or (b) building small screens of FitFolio itself, starting from the proof of concept. Angad and Zaid will build real screens, because the official guides cover the basics and the project's own screens expose the problems we will actually face.

- **Health platform integration:** (a) Apple's and Google's developer documentation, or (b) small test apps that read steps through each bridge library on a physical phone. Preston and Zaid will build the test apps, because reading health data was the main risk in our Development Plan, and only a real device shows how permissions and late data behave.

- **FastAPI and PostgreSQL:** (a) the FastAPI tutorial and PostgreSQL documentation, or (b) applying what we learned in SFWRENG 3DB3 to a first version of our schema, with the backend tests running in CI. Farid and Davis will do both: the tutorial first for FastAPI, which neither has used before, then the schema, because trade consistency has to be designed in from the start.

- **Market data and investing knowledge:** (a) the Finnhub API documentation and its rate limits, or (b) reputable Canadian sources such as the Financial Consumer Agency of Canada. Davis will take the API documentation because he owns the market engine, and Farid will use the Canadian sources because he owns the learning content, which must be correct.

- **Privacy and security:** (a) reading PIPEDA guidance and the OWASP Mobile Application Security Verification Standard, or (b) peer review of each pull request against our safety and security requirements. The whole team will use peer review, because it checks the actual code, with the OWASP standard as the checklist.

- **Usability testing:** (a) the methods from SFWRENG 4HC3, or (b) short pilot sessions with friends before the formal tests. The team will run pilot sessions first, because they are cheap and reveal unclear wording early, then use the 4HC3 methods for the measured tests in the V&V Plan.

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